

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads	)	Docket No. RM26-4-000
to the Interstate Transmission System	)	

**COMMENTS OF EOLIAN L.P.
ON THE ADVANCE NOTICE OF PROPOSED RULEMAKING**

Eolian L.P. (Eolian) respectfully submits comments in response to the Advance Notice of Proposed Rulemaking (ANOPR) on Ensuring the Timely and Orderly Interconnection of Large Loads.¹ As discussed herein, Eolian is a developer of energy generation and load projects, including battery energy storage systems, that develops resources, among other purposes, to serve large loads. Eolian strongly supports the Commission’s effort to ensure the timely and orderly interconnection of large loads to the transmission system.

Eolian urges the Commission to act on the principles in the ANOPR as necessary to ensure just and reasonable and not unduly discriminatory or preferential Commission-jurisdictional rates. Eolian focuses on the critical importance of reforms to enable flexibility through curtailable (or “flexible”) load and curtailable and dispatchable (also “flexible”) hybrid resources. Flexibility commitments² from loads and hybrid resources can help to protect existing customers by avoiding the need for transmission system upgrades and allowing large loads to reduce the resource adequacy impacts of their interconnection and allow large loads—and the resources needed to serve them—to come online quickly without the need for extensive study or

¹ *Ensuring the Timely and Orderly Interconnection of Large Loads*, Advance Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 23, 2025) (ANOPR).

² See Miles Farmer et al., *How DOE’s Proposed Large Load Interconnection Process Could Unlock the Benefits of Load Flexibility*, Nicholas Institute for Energy, Environment & Sustainability, Duke University, at 3-5 (2025), <https://nicholasinstitute.duke.edu/sites/default/files/publications/how-does-proposed-large-load-interconnection-process-could-unlock-benefits-load-flexibility.pdf> (describing large load flexibility commitments and their value).

network upgrades. With the right federal legal framework, the grid can sustain large load growth that is critical to national economic and security interests while reducing the impact of such growth on transmission rates, resource adequacy challenges, and wholesale energy and capacity costs.

The Commission has ample authority to correct these deficits, both via its jurisdiction over interstate transmission and its authority to ensure the justness and reasonableness of practices affecting jurisdictional rates (Section III). Specifically, Eolian supports and provides further reasoning to justify the Commission’s assertion of jurisdiction over large load interconnection. Eolian believes that such assertion is needed to remedy the ways in which existing rates have become unjust and unreasonable by failing to recognize the value that flexibility commitments and hybrid resources provide—and thus inhibiting much needed solutions to current cost and reliability challenges—and failing to prevent opportunities for undue discrimination in large loads’ and hybrid resources’ receipt of interconnection and transmission service. In Section IV, Eolian offers additional rationale and evidence to support a series of reforms that would eliminate barriers to flexibility and provide commercial incentives to invest in sufficiently flexible load and hybrid resources to substantially reduce the impact of large loads on the transmission system and resource adequacy concerns. To remedy these defects, Eolian identifies in Section VI the key elements of reforms to enable enforceable commitments by flexible loads and hybrids, which are also presented in list form in Attachment B. Essential to unlocking this capability and thereby achieving speed-to-market for large loads while safeguarding affordability and reliability are: (1) creation of a standardized large load interconnection process, with a prioritized load interconnection process for flexible (i.e., curtailable, dispatchable) loads; (2) creation of a streamlined, single study, serial interconnection

process for hybrid facilities meeting particular eligibility requirements to enable such qualified hybrids the most rapid speed-to-market; (3) reform of transmission service requirements to enable flexible loads and hybrids to better reflect their use of the transmission system; (4) modification of the standardized generator interconnection process to enable differing study of generation facilities paired operationally to load; (5) modification of Commission-jurisdictional resource adequacy constructs, as necessary, to recognize the contributions of flexible loads and hybrids to resource adequacy; and (6) provision for an interconnection customer option to build. Section VII expands upon one of these key reforms, creation of a streamlined, single study, serial interconnection process (“streamlined study and interconnection process”) for hybrid facilities. The comments, and particularly the streamlined study and interconnection process, are supported by the affidavit of Andrew Levitt and Aniruddh Mohan of the Brattle Group and Ryan Quint and Sirisha Tanneeru of Elevate Consulting in Attachment A.

Finally, Eolian notes that while these reforms are critically necessary and present a significant opportunity to meaningfully address many of the competitiveness, affordability, and reliability challenges the power system and broader economy currently face there is a substantial risk of disruption if the Commission’s entry into this new jurisdictional area is undertaken on too ambitious a timeline or with insufficiently ambitious of a scope. Eolian urges the Commission to act in a timely but sensible timeframe for a rulemaking of this magnitude, importance, and complexity. Moreover, the Commission should provide clear, advanced notice of a jurisdictional transition to enable an orderly transition to new rules and mitigate regulatory uncertainty for affected large loads (Section V).

I. EOLIAN’S INTEREST IN THE RULEMAKING PROCEEDING

Eolian develops energy generation and storage assets throughout the United States and has originated 27 GW of operating capacity to date. Eolian and our partners have developed projects supporting large loads in multiple regions across the United States. Should the Commission issue a final rule on large load interconnections, Eolian expects to participate in transmission providers’ processes developed in compliance with the final rule to develop generation, storage, and load resources.

II. TERMINOLOGY

For reform to be effective, the terms and concepts used in the ANOPR require further definition and specification. In these comments Eolian uses the following terms, and urges the Commission to adopt definitions consistent with the concepts embodied in the following:

- “Flexibility” means the use of flexible load or hybrid facilities;
- “Flexible load” means loads that have made enforceable commitments to respond to curtailment instructions from the transmission provider or balancing authority, as applicable;
- “Hybrid facility” means operationally-paired load and generation that is not limited to co-located load and generation pairs;
- “Flexible hybrid” means a hybrid facility that has made enforceable commitments to respond to curtailment and dispatch instructions from the transmission provider or balancing authority, as applicable; and
- “Expedited,” in the context of interconnection and interconnection studies, means either “streamlined” (that is, requiring less complexity), “prioritized” (that is, addressed earlier-in-time), or both.

Note that a flexible hybrid that can be subject to a streamlined interconnection study process is a narrower class of flexible hybrid facilities, which includes configurations where new supporting generation is electrically close to a new large load, as discussed in Section VII.

III. JURISDICTION

a. The Commission has ample jurisdiction to pursue the ANOPR's reforms

The Commission may assert jurisdiction over the interconnection of large loads to the interstate transmission system. The Federal Power Act vests the Commission with jurisdiction over “the transmission of electric energy in interstate commerce” and “over all facilities for such transmission or sale of electric energy.”³ The Commission’s jurisdiction over transmission does not depend on whether transmission service is part of a wholesale sale.⁴ Rather, all that matters is whether interconnection to the transmission system is a transmission service. It plainly is. Indeed, the interconnection of large loads to the interstate transmission system is a transmission service in precisely the same way that the interconnection of generators to the transmission system is a transmission service, over which the Commission has long asserted jurisdiction.⁵

³ 16 U.S.C. 824(b).

⁴ See *New York v. FERC*, 535 U.S. 1, 16-17 (2002) (“There is no language in the [FPA] limiting FERC’s transmission jurisdiction to the wholesale market.”).

⁵ See *Standardization of Generator Interconnection Agreements & Procs.*, Order No. 2003, 104 FERC ¶ 61,103 (2003), *order on reh’g*, Order No. 2003-A, 106 FERC ¶ 61,220, at P 698 (“interconnection is an essential element of [t]ransmission [s]ervice.”), *order on reh’g*, Order No. 2003-B, 109 FERC ¶ 61,287 (2004), *order on reh’g*, Order No. 2003-C, 111 FERC ¶ 61,401 (2005), *aff’d sub nom. Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 475 F.3d 1277 (D.C. Cir. 2007); *Improvements to Generator Interconnection Procs. & Agreements*, Order No. 2023, 184 FERC ¶ 61,054, *order on reh’g*, 185 FERC ¶ 61,063 (2023), *order on reh’g*, Order No. 2023-A, 186 FERC ¶ 61,199, *errata notice*, 188 FERC ¶ 61,134 (2024); see also *PJM Interconnection, L.L.C.* 190 FERC ¶ 61,115, at P 72 (2025) (“[T]he Commission has exclusive jurisdiction over the provision of transmission service used to serve co-location arrangements, again provided that transmission of electricity in interstate commerce.” (emphasis in original)); *Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 475 F.3d 1277, 1280 (2007) (affirming Order No. 2003 and observing that “interconnections appear to be relationships between parties with respect to electricity flowing over facilities. . . . By establishing standard agreements [the Commission] has exercised its jurisdiction over the terms of those relationships.”); *Pac. Gas & Elec. Co.*, 115 FERC ¶ 61,193, at P 36 (2006) (“[I]nterconnection is part and parcel of transmission of electric energy in interstate commerce, and thus interconnection service is part and parcel of jurisdictional transmission service.”); *id.* P 36 n.52 (citing *Tenn. Power Co.*, 90 FERC ¶ 61,238, at 61,761, *reh’g dismissed*, 91 FERC ¶ 61,271 (2000)).

The fourteen principles advanced by the ANOPR also all fall squarely within the Commission’s jurisdiction. These principles mostly address aspects of the load interconnection process that the Commission has long regulated on the generator side, including: the interconnection study process,⁶ protection systems to govern injections into and withdrawals from the transmission system,⁷ allocation of the cost of network upgrades necessary to facilitate interconnection,⁸ and the interconnection customer’s option to build.⁹ Where the fourteen principles extend beyond interconnection itself, they address subjects that, equally clearly, fall within the Commission’s jurisdiction, including the provision of transmission service and ancillary services to large loads and hybrid resources,¹⁰ and measures to ensure reliability of the bulk power system.¹¹

The policies contemplated in the ANOPR can also steer clear of any interference with state authority over retail electricity sales. The Commission has held that a Commission-jurisdictional tariff may not dictate the terms of retail agreements between large loads and their load-serving entities (terms on service length, minimum “load and energy requirements for retail purchases,” etc.).¹² Nothing in the ANOPR would require such an outcome. To the contrary, the ANOPR focuses exclusively on the relationship between the transmission provider and the large load/interconnection customer, and the Commission can readily limit its action to requirements between those entities while achieving the critical policy goals of the ANOPR. Nothing in the ANOPR speaks to the commercial relationship between the large load and the load-serving

⁶ ANOPR at PP 21 & 24.

⁷ *Id.* P 23.

⁸ *Id.* P 25.

⁹ *Id.* P 26.

¹⁰ *Id.* PP 28–29.

¹¹ *Id.* PP 27 & 31.

¹² *Tri-State Generation and Transmission Association, Inc.*, 193 FERC ¶ 61,070, at PP 49-50 (2025).

entity. Indeed, nothing in the ANOPR speaks directly to energy sales, much less the terms of retail sales.

b. The Commission can avoid jurisdictional limits over distribution facilities, but should establish bright-line rules for which large loads are subject to the ANOPR's policies

The Commission can also assert jurisdiction over the interconnection of large loads to the transmission system as contemplated in the ANOPR without interfering with the jurisdiction “over facilities used in local distribution” that the Federal Power Act has reserved to the States.¹³ In doing so, however, it is important that the Commission establish predictable, workable bright-line rules for which large loads are subject to the policies contemplated by the ANOPR, and which are not.

The ANOPR recognizes that the Commission has established a seven-factor test for distinguishing transmission facilities from distribution facilities and proposes that the Commission use this test to determine which large load interconnections are subject to its jurisdiction “to avoid *even arguably* affecting the States’ jurisdiction over . . . facilities used in local distribution.”¹⁴ Unfortunately, the seven-factor test is no model of clarity, as evidenced by the volume of litigation it has engendered.¹⁵ The ANOPR’s objective of facilitating timely interconnections of new large loads would be frustrated if the rule’s core jurisdictional boundary (distribution vs. transmission) is clouded by unanswered questions or protracted litigation about

¹³ 16 U.S.C. § 824(b)(1).

¹⁴ ANOPR at P 18.

¹⁵ See, e.g., *Midcontinent Indep. Sys. Operator, Inc.*, 181 FERC ¶ 61,056 (2022), *order on reh’g*, 182 FERC ¶ 61,095 (2023); *Sw. Power Pool, Inc.*, 180 FERC ¶ 61,192 (2022); *DTE Electric Company v. Midcontinent Independent System Operator, Inc.*, 180 FERC ¶ 61,222 (2022); *Southern California Edison Co.*, 153 FERC ¶ 61,384 (2015); *ISO New England Inc.*, 178 FERC ¶ 61,115, *order on reh’g*, 179 FERC ¶ 61,186 (2022), *aff’d sub nom. Green Development, LLC v. Federal Energy Regulatory Commission*, 77 F.4th 997 (D.C. Cir. 2023); *Consumers Energy Co. v. Midcontinent Independent System Operator, Inc.*, 171 FERC ¶ 61,020, *order on reh’g*, 172 FERC ¶ 61,201 (2020); *Sw. Power Pool, Inc.*, 149 FERC ¶ 61,051 (2014); *City of Pella, Iowa v. Midwest Independent Transmission System Operator, Inc.*, 134 FERC ¶ 61,081 (2011).

how the seven-factor test will apply. So too would the ANOPR's objective be frustrated by an overly conservative reading of its jurisdictional authority over large load interconnection, as would occur if federally-regulated interconnection service applied only to loads physically connecting to transmission facilities as defined by the seven-factor test. Only one of the seven factors—"local distribution systems will be of reduced voltage"—clearly point toward designating the facilities interconnecting the type of large loads implicated by the ANOPR as transmission, while the other six would tend toward designation of large load interconnection facilities as distribution.¹⁶

To prevent protracted litigation or an overly conservative reading of its authority from obstructing its policy goals, the Commission should establish a bright-line standard over which large loads are subject to a federally-regulated interconnection process instead of relying on the seven-factor test. Eolian recommends the use of a straightforward capacity and voltage threshold. For example, consistent with current engineering convention and Commission precedent,¹⁷ the Commission could establish that the interconnection of large loads (e.g., 20

¹⁶ That is, large loads typically connect to high-voltage facilities that: (1) are in close proximity to the large load customer; (2) are primarily radial in nature; (3) have power flow in but rarely out; (4) do not serve to wheel power to another market; (5) are used for power consumption in a limited geographic area; and (6) have meters at the interface to measure flows into the system. *See Promoting Wholesale Competition Through Open Access Non-Discriminatory Transmission Services by Public Utilities; Recovery of Stranded Costs by Public Utilities and Transmitting Utilities*, Order No. 888, 61 FR 21540, at 50 (May 10, 1996), FERC Stats. & Regs. ¶ 31,036 (1996) (cross-referenced at 75 FERC ¶ 61,080), *order on reh'g*, Order No. 888-A, 62 FR 12274 (Mar. 14, 1997), FERC Stats. & Regs. ¶ 31,048 (cross-referenced at 78 FERC ¶ 61,220), *order on reh'g*, Order No. 888-B, 81 FERC ¶ 61,248 (1997), *order on reh'g*, Order No. 888-C, 82 FERC ¶ 61,046 (1998), *aff'd in relevant part sub nom. Transmission Access Pol'y Study Grp. v. FERC*, 225 F.3d 667 (D.C. Cir. 2000), *aff'd sub nom. N.Y. v. FERC*, 535 U.S. 1 (2002).

¹⁷ *See Order Approving Revisions to North American Electric Reliability Corporation Rules of Procedure and Requiring Compliance Filing*, 187 FERC ¶ 61,196, at PP 39, 55 (2024) (approving NERC use of 20 MVA and 60 kV thresholds as reasonable bright-line rules for inverter based resources that have a material impact on the reliable operation of the Bulk-Power System but are not connected to the distribution system); FERC Staff, Reliability Primer 20 (2020), https://www.ferc.gov/sites/default/files/2020-04/reliability-primer_1.pdf ("The distribution system is generally comprised of sub-100kV lines"); FERC, *Formula Rates in Electric Transmission Proceedings: Key Concepts and How to Participate*, (last updated July 5, 2022), <https://www.ferc.gov/formula-rates-electric-transmission-proceedings-key-concepts-and-how-participate> ("Transmission lines, transformers, substations and other equipment have voltages of 69 kilovolts . . . and above, while distribution lines typically operate at 35 kV or less").

MW, as proposed by the ANOPR) to facilities with voltage greater than or equal to between 60-100 kV would be subject to federal interconnection standards. In its forthcoming Notice of Proposed Rulemaking, the Commission can take comment on the most appropriate nameplate capacity and interconnection voltage thresholds.

Eolian recognizes the ANOPR's first principle and supports the Commission treading carefully into new jurisdictional areas that have traditionally been regulated by states (as discussed more fully below). To the extent that the Commission finds the bright line outlined above would veer too closely to state jurisdictional facilities, Eolian supports the Commission adopting a higher capacity and/or voltage threshold for designation of large loads subject to federal interconnection rules as an approach that would provide certainty without compromising the ANOPR's policy goals. For example, Eolian generally does not develop projects smaller than 75 MW connected at 69 kV, and most loads and hybrid facilities would exceed 100 MW and connect at least at 115 kV.

The Commission can achieve needed clarity without overstepping its jurisdictional authority in either of two ways: (1) by more clearly defining the scope of interstate transmission (as compared to distribution) subject to the Commission's jurisdiction, at least with respect to large-load interconnection, or (2) by establishing interconnection rules for a bright-line set of large loads pursuant to a combination of its jurisdiction over direct transmission interconnection and its "affecting" jurisdiction over certain distribution-connected loads. Either pathway would result in the same bright line, easy-to-apply threshold for whether a large load is subject to Commission's jurisdiction for purpose of interconnection.

First, the Commission could clarify by rule its definition of jurisdictional transmission facilities in the context of large load and hybrid interconnections, and either modify or depart

from its application of the seven factor test. The Federal Power Act does not define transmission facility or distribution facility. Thus, the Commission must inevitably draw a line between the two. Acknowledging that line drawing is never perfect,¹⁸ the Commission can revisit the appropriateness of its seven-factor test for this circumstance and instead adopt a bright line rule, such as the voltage/capacity cutoff proposed above. In its forthcoming Notice of Proposed Rulemaking (NOPR), the Commission should recognize that this would constitute a change in its position and explain the appropriateness of such change, in this limited circumstance, given the importance of a bright-line standard for establishing an administrable, predictable load interconnection process.¹⁹ Such a change in policy is appropriate given that large loads pose different real-time operational reliability and resource adequacy challenges for the transmission system compared to gradual and geographically dispersed load growth.²⁰

Alternatively, the Commission could justify application of a bright-line test for which large loads would be subject to its rules based on a combination of its jurisdiction over transmission facilities and its authority to establish rules, regulations, and practices affecting transmission and transmission rates.²¹ Just as FERC has authority “over generation facilities to

¹⁸ See *American Public Gas Association v. FPC*, 567 F.2d 1016, 1056 (D.C. Cir. 1977) (“An agency confronted with a complex task may rationally turn to simplicity in ground rules, and administrative convenience, at least where no fundamental injustice is wrought.” (quoting *Gulf Oil Corp. v. Hickel*, 435 F.2d 440, 446 (D.C. Cir. 1970))); *Ashland Expl., Inc. v. FERC*, 631 F.2d 817, 822 (D.C. Cir. 1980) (“It is familiar doctrine that reviewing courts should accommodate agency flexibility to permit adjustments based on exigencies of administration.” (citations omitted)); *Gulf Oil Corp. v. Hickel*, 435 F.2d at 446 (“Administrative agencies and officers have a primary task to administer broad policy mandates for the common good of our society, and they cannot be required to refine their rules to assure tailor-made equity for each of the complexities that may arise. If they issue and interpret regulations which are rational and supportable in their general application, they cannot necessarily be charged with unreasonableness because in particular applications the regulations and general interpretations may grind with a rough edge.”).

¹⁹ See *FCC v. Fox Television Stations, Inc.*, 556 U.S. 502, 515 (2009); *Baltimore Gas & Elec. Co. v. FERC*, 954 F.3d 279, 286 (D.C. Cir. 2020).

²⁰ See e.g., Southwest Power Pool, Filing, Docket No. ER26-247-000, Transmittal Letter at 9-10 (filed Oct. 24, 2025) (“The Challenges Presented by Large Loads”) (“Large loads may exhibit non-conforming load profiles, the ability to ramp rapidly [] and flutter, and sensitivity to power quality issues. The impact of these operational characteristics is magnified because large loads are also concentrated at a single site.”)

²¹ 16 U.S.C. §§ 824d(a), 824e(a).

the extent necessary to regulate interstate transmission,”²² so too does FERC have authority to prescribe rules for interconnections at certain voltage thresholds to the extent necessary to regulate interstate transmission. Large loads interconnecting above a high voltage threshold will directly affect transmission system reliability, regardless of whether those loads connect directly to the transmission system or to the rare distribution facility operating at such voltage.²³

Importantly, were FERC to assert jurisdiction under this second option, it could establish a bright line for when a federal rather than state load interconnection process applies, without making a determination as to whether the facility to which the load is connecting is transmission or distribution. This approach would eliminate the necessity of applying the seven-factor test as a preliminary step to determine whether a large load interconnection is Commission-jurisdictional as a preliminary step to determine whether a large load interconnection is Commission-juris. Subjecting large load interconnections to Commission rules on the basis that they directly affect jurisdictional rates would not necessarily mean that the relevant load is interconnecting to a transmission facility, because the bright line under such a theory would be based on the interconnection’s effects on jurisdictional rates, not the status of the facility. Accordingly, adopting this approach to FERC jurisdiction over large load interconnection would not alter the underlying rate treatment of any facilities that may currently be classified as distribution.

Just as FERC is on solid jurisdictional grounds to “exercis[e] jurisdiction . . . over interconnections to a ‘distribution’ facility when the facility is included in a public utility’s Commission-filed OATT and the interconnection is for the purpose of facilitating a jurisdictional

²² See *Transmission Access Pol’y Study Grp. v. FERC*, 225 F.3d 667, 683 (D.C. Cir. 2000).

²³ *Supra* n.20. The characteristics of large loads that pose operational reliability and resource adequacy concerns persist even if interconnected at lower voltage facilities, particularly given the potential for concentrated impact of large load in a localized area and rapid growth in the additions of large loads.

wholesale sale of electric energy,”²⁴ FERC has authority to exercise jurisdiction over interconnections made to facilitate transmission service. For the reasons described below, establishing rules for large load interconnection—and in particular hybrid facility interconnection—will significantly facilitate just and reasonable wholesale rates. FERC’s “affecting” jurisdiction is sufficient to enable it to set bright-line rules in such circumstances.²⁵

This approach would be consistent with the Southwest Power Pool’s (SPP) recently High Impact Large Load (HILL) proposal.²⁶ There, SPP proposes to establish specific interconnection procedures for HILLs on the basis that “a facility requiring a large amount of energy at a single site . . . can present operation challenges and reliability risks to the Transmission System.”²⁷ SPP proposes to define HILL as loads “either (1) 10 MW or more if connected to the Transmission System at a voltage level less than or equal to 69 kV; or (2) 50 MW or more if connected to the Transmission System at a voltage level greater than 69 kV.”²⁸ SPP explains that this definition is necessary because, “[w]hile the NERC definition of the Bulk Electric System includes only transmission elements of 100 kV or higher, . . . a HILL connecting to transmission facilities at or below 69 kV will have similar impacts to the Transmission System as a HILL that is 50 MW, or larger, connecting to transmission facilities that are greater than 69 kV.”²⁹ While the Commission need not adopt these exact thresholds, it can take a similar approach for establishing bright-line rules for the interconnection of large loads in service of the policy objectives of the ANOPR and the need for reform as outlined below.

²⁴ *Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 475 F.3d 1277, 1282 (D.C. Cir. 2007) (citations and quotations omitted).

²⁵ *See NextEra Energy Res., LLC v. FERC*, 118 F.4th 361, 368-69 (D.C. Cir. 2024) (finding FERC jurisdiction to establish rules regarding a circuit breaker at a generating facility because of the impacts of such circuit breaker on the transmission system).

²⁶ SPP, Filing, Docket No. ER26-247-000, Transmittal Letter at 12 (filed Oct. 24, 2025).

²⁷ *Id.*

²⁸ *Id.* at 12 & Proposed Tariff, Part I, Section 1 (Definitions H) (Definition of High Impact Large Load).

²⁹ *Id.* at 13.

It is important to note that this is a limited authority that does not involve FERC asserting jurisdiction over the distribution facilities *per se*. Rather, under this approach, FERC would assert jurisdiction only over the terms and conditions of interconnection of large loads because such jurisdiction is necessary to ensure the justness and reasonableness of jurisdictional rates.³⁰ Such an approach is consistent with the Federal Power Act’s jurisdictional boundaries, just as FERC’s regulation of generator interconnection processes and wholesale electricity market rules does not constitute an impermissible exercise of jurisdiction “over facilities used for the generation of electric energy”³¹ and for the same reason FERC’s regulation of wholesale demand response “does not run afoul of § 824(b)’s proscription” against federal regulation of “the quantity or terms of retail sales.”³²

IV. NEED FOR REFORM

Having asserted jurisdiction, FERC has substantial reason and evidence to establish uniform rules for large load interconnection. The status quo has become unsustainable. The speed, scope, and operational characteristics of large load additions mismatch with legacy systems in a way that risks rising costs, reliability challenges, and squandered economic development and national security opportunities. Existing tariff provisions prevent the solutions needed to cut through this morass, and have therefore become unjust and unreasonable. In addition, the same concerns of undue discrimination that motivated the Commission to adopt the *pro forma* open access transmission tariff and *pro forma* generator interconnection procedures

³⁰ See *Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 475 F.3d 1277, 1282 (D.C. Cir. 2007) (concluding that FERC did not err “by not applying its seven-factor test . . . to determine whether a facility is an exempt local distribution facility or a covered transmission facility” because “FERC is exerting jurisdiction over transactions, based on the transactions’ satisfaction of the Act’s jurisdictional criteria”)

³¹ 16 U.S.C. 824(b)(1).

³² *FERC v. Elec. Power Supply Ass’n*, 577 U.S. 260, 281 (2016).

also apply in the large load interconnection context, and thus require adoption of *pro forma* large load interconnection procedures.

a. Existing tariff provisions that prevent the recognition and facilitation of large load flexibility commitments have become unjust and unreasonable

First, the large load interconnection process risks replicating the challenges that have plagued the generator interconnection process. In generator interconnection processes, a dramatic increase in the number of interconnection requests and limited transmission capacity have caused interconnection queue backlogs, longer development timelines, and increased uncertainty regarding the cost and timing of interconnecting to the transmission system.³³ The median generation project now takes roughly five years to enter commercial operation from the time it submits an interconnection request.³⁴ Similar delays have begun to appear in the large load interconnection process, with reports that data centers face a seven year delay to interconnect to the Dominion system.³⁵ These delays are already having an effect; SPP, for example, noted in a recent filing that large load interconnection requests in SPP peaked in 2022, potentially “due in part to SPP’s inability under current processes to study and interconnect large loads quickly enough to satisfy commercial needs.”³⁶ The Department of Energy casts the potential for slow large load interconnection to constrain data center development in stark terms:

³³ Order No. 2023, 184 FERC ¶ 61,054 at PP 37-38.

³⁴ Joseph Rand et al., *Queued Up: 2024 Edition: Characteristics of Power Plants Seeking Transmission Interconnection as of the End of 2023*, Lawrence Berkeley National Laboratory, at 41 (Apr. 2024), available at https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_R2.pdf.

³⁵ Josh Saul, *Data Centers Face Seven-Year Wait for Dominion Power Hookups*, BLOOMBERG (Aug. 29, 2024), <https://www.bloomberg.com/news/articles/2024-08-29/data-centers-face-seven-year-wait-for-power-hookups-in-virginia?sref=B2BBHw9t>; see also Jeff St. John, *One Way Data Centers Can Help the Grid? By Being Flexible*, CANARY MEDIA (Feb. 27, 2025), <https://www.canarymedia.com/articles/utilities/one-way-data-centers-can-help-the-grid-by-being-flexible> (reporting that, according to one utility executive, the backlog for connecting large data centers to the grid is now at least five years); Tom Kleckner, *Large Loads Slow to Interconnect in ERCOT*, RTO INSIDER (Oct. 27, 2025), <https://www.rtoinsider.com/118099-ercot-stakeholders-talk-large-loads/> (noting that, of the 130 GW of large load interconnection requests in ERCOT over the past 10 months, “only about 6.5 GW have been energized or approved for energization, with an additional 4.7 GW being studied.”).

³⁶ SPP, Filing, Transmittal Letter, Docket No. ER26-247-000, at 8 (filed Oct. 24, 2025).

“A failure to power the data centers needed to win the AI arms race or to build the grid infrastructure that ensures our energy independence could result in adversary nations shaping digital norms and controlling digital infrastructure, thereby jeopardizing U.S. economic and national security.”³⁷ Second, large load additions are requiring significant transmission system additions at a time of already rising transmission system costs.³⁸ Specifically, proliferation of large load interconnection requests in the face of limited transmission system load serving capacity will require increasingly extensive network upgrades and local transmission additions to serve new loads without transmission system performance violations.³⁹ Because local transmission and network upgrade costs identified in load interconnection processes are often recovered through existing transmission rates,⁴⁰ existing customers are paying for the upgrades needed for large loads to be served.⁴¹ In addition to causing an unjust cost shift, this blunts price

³⁷ US Department of Energy, *Resource Adequacy Report: Evaluating the Reliability and Security of the United States Electric Grid*, at 1 (July 2025), <https://www.energy.gov/sites/default/files/2025-07/DOE%20Final%20EO%20Report%20%28FINAL%20JULY%207%29.pdf>.

³⁸ See, e.g., U.S. Energy Information Administration, *Utilities Continue to Increase Spending on the Electric Transmission System*, (Mar. 26, 2021), <https://www.eia.gov/todayinenergy/detail.php?id=47316> (finding that “[a]nnual spending by major U.S. electric utilities on the U.S. electric transmission system has increased from \$9.1 billion (2019 dollars) in 2000 to \$40.0 billion in 2019.”); Claire Wayner et al., *Mind the Regulatory Gap: How to Enhance Local Transmission Oversight*, at 13 (Nov. 2024), <https://rmi.org/insight/mind-the-regulatory-gap/> (finding that “the share of customer bills represented by transmission and distribution costs has been increasing steadily, from 10% in 2005 to 24% in 2020.”).

³⁹ Ryan Quint et al., *Practical Guidance and Considerations for Large Load Interconnections*, ELEVATE ENERGY CONSULTING & GRIDLAB, at 14 (May 2025), <https://gridlab.org/portfolio-item/practical-guidance-and-considerations-for-large-load-interconnections/> (“As more large loads seek connection at higher voltage levels, it can be assumed that the cost of network upgrades will also increase. This is because the cost of transformers, circuit breakers, switchgears, and other interconnecting network devices increases by voltage class. In addition, larger loads may cause BPS performance violations (thermal overloads, voltage issues, instability conditions, oscillations, or other risks) that must be mitigated to reliably serve these loads.”).

⁴⁰ See PJM, Answer, Transmittal Letter, Docket Nos. EL25-49-000, AD24-11-000, EL25-20-000, at 7 (Mar. 24, 2025) (noting that solutions to needs identified in studies “to identify local upgrades needed to reliably accommodate [Network Loads] based on local transmission owner planning criteria” “typically take the form of Supplemental Projects” that “are cost allocated to the local Transmission Owner’s zone pursuant to individual Transmission Owner cost allocation frameworks.”).

⁴¹ Mike Jacobs, *Connection Costs: Loophole Costs Customers Over \$4 Billion to Connect Data Centers to Power Grid*, UNION OF CONCERNED SCIENTISTS, at 1 (Sept. 2025), <https://www.ucs.org/sites/default/files/2025-09/PJM%20Data%20Center%20Issue%20Brief%20-%20Sep%202025.pdf> (“in 2024 alone, utilities in seven PJM states passed more than \$4.3 billion in additional costs on to customers, with billions more still to come.”); Eliza Martin & Ari Peskoe, *Extracting Profits from the Public: How Ratepayers Are Paying for Big Tech’s Power*,

signals to encourage large loads to select sites that minimize transmission system costs. What is more, the need for network upgrades, and more extensive network upgrades, further slows large load interconnection schedules because network upgrades often take several years to build. The recent PJM Transition Cycle 2 (TC2) Phase 1 study results are a red flag for the region's ability to add adequate amounts of new capacity quickly given standard planning and operational approaches: while some projects do have zero or minimal network upgrade costs, a significant amount of capacity, including from Reliability Resource Initiative projects, is facing costs of hundreds of dollars per kilowatt of capacity and network upgrade completion timelines of up to 90 months.⁴² As discussed in Eolian's comments on the June Resource Adequacy Technical Conference, Docket AD25-7, Eolian has copious experience with duplicative upgrades and lengthy upgrade timelines and has waited as long as 42 months for a network upgrade.

Third, the rapid addition of large loads is also straining existing resource adequacy processes. Reserve margins are already tight, and marginal capacity prices in both RTO and non-RTO regions have soared.⁴³ Meanwhile, delays in the generator interconnection process prevent new supply from responding to high prices.⁴⁴ If large loads are permitted to interconnect

HARVARD LAW SCHOOL ENVIRONMENTAL & ENERGY LAW PROGRAM, at 15-16 (Mar. 2025),

<https://eelp.law.harvard.edu/wp-content/uploads/2025/03/Harvard-ELI-Extracting-Profits-from-the-Public.pdf>.

⁴² See PJM, *Transition Cycle 2: New Service Requests System Impact Study Executive Summary Report*, (Oct. 29, 2025), https://www.pjm.com/pjmfiles/pub/planning/project-queues/Cluster-Reports/TC2/TC2_PH1_Executive_Summary.htm.

⁴³ See, e.g., Travis Kavulla, Opening Statement of Travis Kavulla, Vice President of Regulatory Affairs, NRG Energy, Inc., Docket No. AD25-7-000, at 4 (filed June 4, 2025) (finding marginal capacity prices ranging from \$118-466/MW across various RTOs and investor owned utility territories); see also Ethan Howland, *PJM Capacity Prices Set Another Record with 22% Jump*, UTILITY DIVE (July 23, 2025), <https://www.utilitydive.com/news/pjm-interconnection-capacity-auction-prices/753798/>; Ethan Howland, *MISO Summer Capacity Prices Jump to \$666.50/MW-day as Power Supplies Shrink*, Utility Dive (Apr. 29, 2025), available at <https://www.utilitydive.com/news/miso-capacity-auction/746576/>.

⁴⁴ See Joseph Rand et al., *Queued Up: 2024 Edition: Characteristics of Power Plants Seeking Transmission Interconnection as of the End of 2023*, Lawrence Berkeley National Laboratory, at 34-35 (Apr. 2024), available at https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_R2.pdf (showing persistently long timelines between interconnection requests and interconnection agreements in FERC-jurisdictional queues across the country).

faster than new supply can keep pace, the result will be reliability risks and increased costs for customers.⁴⁵ Some regions may mitigate reliability challenges by requiring transmission customers to demonstrate that new load additions are matched by commensurate Network Resources.⁴⁶ But while these regions may face reduced reliability impacts, their requirements, coupled with a failure to allow customers to demonstrate through curtailment or dispatch commitments that they have sufficient resources to cover the new loads, have prevented large load customers from being able to timely interconnect to the system.⁴⁷

All of these problems are made worse by the absence of standardized large load interconnection procedures. Eolian has experience across jurisdictions with load interconnection as well as the proposed interconnection of hybrid resources. In Eolian's experience, existing interconnection processes for large loads and hybrid resources suffer from a lack of standardization—both because they differ markedly from jurisdiction to jurisdiction, and because they often rely on unwritten, non-transparent practices and afford unwarranted discretion to the local utility. The absence of standardized, transparent procedures creates opportunities for undue discrimination in queue processing and transmission upgrade cost assignment as public utilities negotiate individual contracts with loads. Further, the absence of standardized provisions to

⁴⁵ See, e.g., Governor Josh Shapiro and the Commonwealth of Pennsylvania, Complaint, Docket No. EL25-49-000, at 1 (filed Dec. 30, 2024) (“Pennsylvania ratepayers face potentially the largest unjust wealth transfer in the history of U.S. energy markets”); see also *id.* at 23 (“[S]etting the price cap at Gross CONE would likely increase capacity prices for the 2026/2027 BRA by as much as 50% relative to prices under a Net CONE-based price cap, with no reasonable expectation of incremental market response sufficient to justify this cost. This would represent an unjustified wealth transfer as the incremental capacity and reliability benefit are shown to be minimal and come at cost orders of magnitude greater than any reasonable estimate of the VOLL.”).

⁴⁶ See, e.g., Charlton Hill, *Overview of Network Integration Transmission Service Requirements*, SPP, at 12 (Feb 9, 2024), <https://spp.org/Documents/71088/TWG%20Attachment%20AQ%20Focus%20Group%20Materials%2020240209.zip> (taking the position that “the Network Customer must have enough Network Resources to serve their Network Load.”).

⁴⁷ See SPP, Filing, Transmittal Letter, Docket No. ER25-2430-000, at 11 (filed June 4, 2025) (noting that SPP has denied “many transmission customers[’]” load study requests because the transmission customers lacked sufficient designated resources.”).

deter or weed out speculative interconnection requests risks bogging down load interconnection study processes just as the nation has experienced with generator interconnection.

Fortunately, a technically available solution could cut through this Gordian Knot: recognizing enforceable commitments from large load customers and hybrid facilities to curtail and dispatch when called upon, providing transmission service and balancing authority operations protocols that match these commitments, and appropriately rewarding these commitments with preferred treatment in the interconnection process. With the right framework of federal requirements, integrating the interconnection processes for load and electrically close generation resources and providing for study and service based on only the withdrawal rights requested, as the ANOPR contemplates, could help to expedite study timelines, minimize the need for network upgrades, and provide an incentive and efficient path for large loads to bring their own generation, neutralizing effects on broader resource adequacy.

It is obvious from basic engineering principles that generation located near load requires less transmission infrastructure to serve that load than generation located distant from load. Existing rules, however, do not allow developers to *intentionally* capture the benefits of proximity between load and generation through coordinated additions. While planned generators may be studied in relation to load that *already exists* (and vice versa for planned loads), integrated study of planned large loads and resources developed exclusively to serve them is almost never permitted. Eolian has explicitly requested integrated study of load and proximate generation in multiple jurisdictions, including generation developed near publicly-announced load developments as well as generation that Eolian (or its development partners) are developing near neighboring load that Eolian *is also developing*. In Eolian's experience, grid operators normally do not accommodate these requests either due to their understanding of state

jurisdictional rules and planning standards (thus demonstrating the need for clarity at the Federal level) or because of the commercial risk that either the load or the generation project may be delayed or cancelled. While Eolian agrees that all development projects face risks to completion, ultimately cancellation of only one half of a coordinated load/generation development would introduce commercial pressure to develop the corresponding resource—thereby ultimately resolving itself in a functioning market or planning regime. Grid operators should be focused on maintaining reliability during such transitory conditions, including by allowing load flexibility on the part of large loads.

By contrast, Eolian has experienced multiple cases where a transmission provider's refusal to perform an integrated study of load and generation has caused duplicative upgrade requirements that could have been avoided if load and generation had been considered together, which has impacted the business viability of both load and generation. In addition, the absence of non-firm network transmission service has left large loads without the option to be served by the grid during normal conditions but to agree to curtail (in whole or in part) during times of transmission system or resource adequacy stress (including stress conditions that would be resolved by development of additional generation resources). Similarly, existing interconnection processes lack an effective pathway to recognize and reward large loads that have made firm commitments to operate flexibly, and balancing authorities lack mechanisms to provide curtailment and dispatch instructions to transmission customers based on commitments that such customers have made as part of the interconnection process. This causes balancing authorities to procure more capacity than would be necessary were large loads' willingness to curtail in whole or in part accommodated, and drives up costs for wholesale electricity customers.

It is therefore clear that current tariff requirements are preventing adoption of the solutions needed to allow loads to timely connect, control rising transmission costs, and address the resource adequacy challenges that large load additions threaten to present. Because interconnection processes for large load customers and hybrid facilities are demonstrably unjust and unreasonable, FERC must now exercise its jurisdiction over these processes to remedy the unjust and unreasonable rates.

b. Existing requirements permit undue discrimination in the large load interconnection process and require adoption of pro forma large load interconnection procedures

In addition, current interconnection and transmission service requirements allow opportunities for undue discrimination. The *pro forma* generator interconnection procedures and *pro forma* OATT must be revised, and *pro forma* large load interconnection procedures adopted, to ensure that large loads seeking to meet their own capacity needs via hybrid arrangements with contracted generation are not subject to undue discrimination.

In Order No. 2003, the Commission explained that “interconnection is a critical component of open access transmission service and thus is subject to the requirement that utilities offer comparable service under the [*pro forma* Open Access Transmission Tariff].”⁴⁸ The Commission found that it was appropriate to establish a standard set of generator interconnection procedures to “minimize opportunities for undue discrimination and expedite the development of new generation, while protecting reliability and ensuring that rates are just and reasonable.”⁴⁹ Specifically, among other benefits, the Commission found that “standard interconnection procedures and a standard agreement” would “(1) limit opportunities for Transmission Providers to favor their own generation, (2) facilitate market entry for generation

⁴⁸ Order No. 2003, 104 FERC ¶ 61,103 at P 9 (citing *Tennessee Power Co.*, 90 FERC ¶ 61,238 (2002)).

⁴⁹ *Id.* P 11.

competitors by reducing interconnection costs and time, and (3) encourage needed investment in generator and transmission infrastructure.”⁵⁰ To the extent that large loads seek to bring their own generation resources to enable flexibility commitments and/or avoid resource adequacy issues, the same principles apply here. That is, there is a need for standardized large load interconnection procedures and/or generator interconnection procedure revisions to “limit opportunities for Transmission Providers to favor their own generation” to serve the new large loads and otherwise facilitate generation competitor market entry and investment in generator and transmission infrastructure.

Similarly, in Order No. 888 the Commission explained that “[n]on-discriminatory open access to transmission services is critical to the full development of competitive wholesale generation markets and the lower consumer prices achievable through such competition.”⁵¹ The Commission therefore concluded that “[d]enials of access (whether they are blatant or subtle), and the potential for future denials of access, require the Commission to revisit and reform its regulation of transmission in interstate commerce.”⁵² As when the Commission first issued Order No. 888, transmission providers’ control of transmission access, whether blatant or subtle, creates opportunities for undue discrimination by transmission providers to connecting large loads and/or the hybrid generators designated to serve them—including in the processing of large load interconnection requests and in the rates and terms through which the large loads agree to take service. This too requires standardized provisions for providing transmission service to large loads in order to stamp out opportunities for undue discrimination.

⁵⁰ *Id.* P 12.

⁵¹ Order No. 888, 61 FR 21540 at 50.

⁵² *Id.*

V. TRANSITION TO COMMISSION JURISDICTION

The ANOPR outlines a rulemaking process that will raise numerous novel and complex issues. Resolving those issues carefully will inevitably take time, as will the compliance process, which typically entails years for major regulatory reforms. Transmission providers may need time to adapt to federal requirements, and to expand the human and other resources necessary to expeditiously process rapidly growing queues of large load interconnection requests. While completing the rulemaking process quickly is important, perhaps even more important is for the Commission to articulate a clear transition rule that prevents uncertainty from slowing down current market progress. To avoid that outcome, the Commission should state as clearly as possible in any forthcoming NOPR (or before) that it intends to provide for a transition rule and how it will propose to handle large load interconnections that have begun but not completed an existing state-jurisdictional process at the time the relevant tariff provisions go into effect. The Commission will need to answer two principal questions:

- (1) For loads that wish to complete the state-jurisdictional process, how far into that process will they need to have progressed before the relevant Commission-jurisdictional tariff provisions go into effect in order to remain subject to the state-jurisdictional process? Is it enough to have submitted an application to the state-jurisdictional utility for interconnection service or must they be under study or already have an interconnection agreement in place?
- (2) Will loads and hybrid resources that have begun the state-jurisdictional process be allowed to switch to the Commission-jurisdictional process if that is their preference?

In the interest of avoiding a regulatory gap in which state regulators lack the authority to process and oversee the interconnection of loads but transmission providers have not yet filed,

received Commission approval, or established the capability to implement federally-regulated load interconnection processes, the Commission should clearly state that state regulators retain jurisdiction to continue to process new load interconnection requests until, at the earliest, an initial compliance filing by the relevant transmission provider is accepted by the Commission and the transmission provider's new load interconnection process has become operative. Any load interconnection customer that has commenced a state jurisdictional process consistent with such a rule should then have the option to continue to be processed pursuant to that process even after acceptance of the applicable compliance filing, in order to limit disruption to the orderly processing of large load interconnections.

The Commission has legal authority to provide for such a jurisdictional transition period. As load interconnection is a matter that has been subject to either federal or state jurisdiction depending on the region, the Commission may appropriately define the point at which federal requirements over load interconnection are sufficiently defined and in effect so as to directly conflict with state jurisdictional requirements.⁵³ The Commission has previously elected to decline to assert jurisdiction over matters that are arguably within the scope of its statutory authority, and has provided for periods of transition where the Commission announces a new jurisdictional line between federal and state authority.⁵⁴

⁵³ See *Miss. Power & Light Co. v. Miss. ex rel. Moore*, 487 U.S. 354, 377 (1988) (“a state agency’s ‘efforts to regulate commerce must fall when they conflict with or interfere with federal authority over the same activity.’” (citation omitted)); *Nantahala Power & Light Co. v. Thornburg*, 476 U.S. 953, 953 (1986) (holding that a state could not use its retail-ratemaking authority to prevent recovery of a FERC-approved rate because to do so would “conflict[] with FERC’s orders.”); see also *Oneok, Inc. v. Learjet, Inc.*, 575 U.S. 373, 388, (2015) (“*Mississippi Power*, however, is best read as a conflict pre-emption case, not a field pre-emption case.”); cf. *New York v. FERC*, 535 U.S. 1, 25, (2002) (“FERC chose not to assert such jurisdiction, but it did not hold itself powerless to claim jurisdiction. Indeed, FERC explicitly reserved decision on the jurisdictional issue”)

⁵⁴ See e.g., *Participation of Distributed Energy Res. Aggregations in Markets Operated by Reg’l Transmission Organizations & Indep. Sys. Operators*, Order No. 2222, 172 FERC ¶ 61,247, 62,661 at PP 101-103 (2020) (“while we are herein declining to exercise jurisdiction over the interconnections of distributed energy resources to distribution facilities for the purpose of participating in distributed energy resource aggregations, the Commission may revisit this policy decision in the future we recognize that such distributed energy resources may already

VI. KEY ELEMENTS OF A RULE TO ENABLE RAPID, AFFORDABLE INTERCONNECTION OF LARGE LOADS

To achieve the significant promise of the ANOPR to enable large load to interconnect and enter into operation expeditiously, a series of reforms are necessary to create and incentivize pathways for large loads that reduce their use of existing transmission and generation capacity, because access to each are key bottlenecks to rapid, affordable development of large loads.

While broader reforms may be warranted to address the multiple goals of the ANOPR, this comment focuses on the essential elements of a rule to eliminate barriers and appropriately incent the “flexible” loads and hybrids (i.e., curtailable, dispatchable) that reduce the need for additional transmission or generation capacity and therefore can be rapidly developed.

This section develops key concepts and requirements that the Commission will need to establish to achieve that goal. We recognize that these changes would need to be made to different sets of tariff provisions, that have traditionally been subject to separate regulatory requirements and Commission precedent. Achieving the aims of the ANOPR, however, depends on reforming the components of load interconnection processes, generator interconnection processes, transmission service requirements, and Commission-jurisdictional resource adequacy constructs that must work together in order to enable flexible loads and hybrids to safely and

have interconnected pursuant to procedures that were accepted by the Commission prior to the effective date of this final rule. Therefore, to minimize disruption to existing interconnection agreements for distributed energy resources, we are not requiring distributed energy resources that already interconnected under Commission-jurisdictional procedures to convert to state or local interconnection agreements.”); *Advanced Energy Econ.*, 161 FERC ¶ 61,245 at P 62 (2017) (recognizing that although the Commission granted retail authorities the ability to “opt-out” demand response resources from participation in wholesale markets in Orders No. 719 and 745, “the Commission was not obligated to do so” and did so as a matter of Commission discretion); *id.* at P 66 (although provision of such an opt-out was not legally necessary, the Commission grandfathered Kentucky’s ability to continue to opt-out energy efficiency resources because it predated the Commission jurisdictional determination).; *cf. Sw. Power Pool, Inc.*, 190 FERC ¶ 61,169 at P 127 (2025) (acknowledging SPP’s commitment to work on state concerns regarding a transition from generator interconnection processes that reflect state procurement policy to the SPP-managed queue).

reliably operate while appropriately avoiding the expense and delay of unnecessary infrastructure investment. Failure to truly enable flexible loads and provide them a means to internalize the benefits they provide to the system could otherwise result in the worst of all worlds: shifting large load interconnection to a standardized federal process that *slows down* progress by disrupting the potential for state innovation and failing to reduce the monumental cost of transmission expansion and capacity growth otherwise necessary to integrate large loads into the transmission system.

As set forth in Attachment B, this comment envisions the following sets of tariff reforms would be required to implement the concepts discussed herein:

- Creation of a standardized large load interconnection process, with a prioritized load interconnection process (i.e., “**queue priority**”) for flexible (i.e., curtailable, dispatchable) loads;
- Creation of a streamlined, single study, serial interconnection process (“**streamlined study and interconnection**”) for hybrid facilities meeting particular eligibility requirements, to enable such qualified hybrids the most rapid speed-to-market. These reforms offer the most potent opportunity to realize the promise of the ANOPR, and are discussed at greater length in Section VII and Attachment A, Testimony of Mr. Levitt, Dr. Mohan, Dr. Quint, and Ms. Tanneeru;
- Reforms to transmission service requirements to enable flexible loads and hybrids to better reflect their use of the transmission system;
- Minor modifications to the standardized generator interconnection process to enable differing study of generation facilities paired operationally to load (but which do not meet the more particular eligibility criteria for study in the streamlined, single study, serial interconnection process for hybrid facilities);
- Modifications to Commission-jurisdictional resource adequacy constructs, as necessary, to recognize the contributions of flexible loads and hybrids to resource adequacy; and
- Provision of an option to build for interconnection customers that improves upon that currently afforded to generator interconnection customers.

a. Rules to grant **queue priority** to curtailable loads and curtailable and dispatchable hybrid facilities

The ANOPR acknowledges the value that flexible loads and dispatchable hybrid resources can offer the grid. The ANOPR proposes that hybrid resources be studied jointly and that those joint studies “minimize the need for costly network upgrades.”⁵⁵ The ANOPR also proposes that, when an interconnecting resource commits to operate flexibly, and when such commitment is “sufficient for the system operator to integrate the facility into both operations and system planning,” the interconnecting study “should be expedited.”⁵⁶ Expediting the study and interconnection of flexible hybrid resources can be accomplished through prioritizing studies and interconnection of those resources, streamlining the process of studying those resources, or both.

To fulfill these concepts advanced in the ANOPR, the Commission should direct transmission providers to provide for a FERC-jurisdictional load interconnection process that grants queue priority to loads that make commitments to curtail, and hybrid facilities that make commitments to dispatch and curtail, such that (a) the load or hybrid resource reduces the system’s need for capacity or Network Resources, and/or (b) avoids the need for network upgrades. A FERC-jurisdictional load interconnection process is central to providing incentive to flexible loads and hybrid facilities. Speed-to-market is a key commercial inducement to incorporating flexibility into large load and hybrid facilities, but requires some degree of standardized rules affording queue priority in return for investment in flexibility. At the same time, such flexibility must be in the form of meaningful commitments to result in reduced transmission system upgrades or avoided capacity procurement. A FERC-jurisdictional load interconnection process can provide for such standardized rules around queue priority, while ensuring that flexibility commitments are appropriately defined, enforced, and recognized in

⁵⁵ ANOPR at P 20.

⁵⁶ *Id.* P 24.

other FERC-jurisdictional processes. In establishing rules to grant queue priority, the Commission must address, at minimum (i) What definition of a hybrid facility should apply for purpose of queue priority? (ii) what curtailment and dispatch commitments are required in order to qualify for priority? (iii) how will such commitments be enforced? and (iv) what does queue priority mean in practice, within the context of the load interconnection process that the Commission may establish, as well as any modifications to the generation interconnection process the Commission may require?

(i) What definition of a hybrid facility should apply for purpose of queue priority?

As the ANOPR appears to recognize, hybrid resources are key to achieving faster interconnection. As discussed below, a hybrid pairing of dispatchable, curtailable load and supply enable speed-to-market by enabling avoidance of network upgrades, centralized capacity procurements, or both, and potentially facilitating faster interconnection studies for both load and supply. But to achieve that goal, the rules governing hybrid resource interconnections should be structured to enable the broadest possible market competition that achieves the relevant objective (avoiding network upgrades; centralized capacity procurement; faster study).

First, the Commission should ensure that the hybrid resource category is technologically neutral, and specifically that it is open to storage resources as well as generators. Storage resources can deploy more quickly than any other grid resource⁵⁷ and are just as capable, if not more capable, of managing the impact of new large loads on the grid. As explained further

⁵⁷ See Joseph Rand et al., *Queued Up: 2024 Edition: Characteristics of Power Plants Seeking Transmission Interconnection As of the End of 2023*, LAWRENCE BERKELEY NATIONAL LABORATORY, at 36, 40 (Apr. 2024), https://emp.lbl.gov/sites/default/files/2024-04/Queued%20Up%202024%20Edition_R2.pdf (finding that, on average, batteries are the fastest technology to study and to reach commercial operations after signing interconnection agreements).

below, when included in a hybrid resource, storage should be permitted to charge through a non-firm or conditional firm transmission service.⁵⁸

Second, to foster competition, the rules governing hybrid resources should allow for the greatest possible diversity in business models and technical configurations. So long as the hybrid resource is technically capable of fulfilling its dispatch commitments, it should not matter whether the generation/storage component of the hybrid resource is located in front of the meter, behind the meter, or at an electrically close point on the grid. The position of the meter matters for purposes of retail service, of course. But it should not matter for purposes of interconnection to the transmission system. For example, the SPP proposes to allow a resource to be studied as serving a specific large load if it is located within two substations of the large load.⁵⁹ Likewise, it should not matter whether distinct elements within the hybrid resources are developed, owned or operated by different entities. So long as the entities that comprise the hybrid resource are able to fulfill their obligations, including enforceable flexibility (i.e., dispatchability and curtailability) commitments, collectively, it should not matter how they divide responsibility among themselves.

⁵⁸ In addition, consistent with Order No. 841, energy storage resources “should not be charged transmission charges when they are dispatched by an RTO/ISO to provide a service.” *Electric Storage Participation in Markets Operated by Regional Transmission Organizations and Independent System Operators*, Order No. 841, 162 FERC ¶ 61,127, at P 293 (2018), *order on reh’g*, Order No. 841-A, 167 FERC ¶ 61,154 (2019), *aff’d sub nom. Nat’l Ass’n of Regul. Util. Comm’rs v. FERC*, 964 F.3d 1177 (D.C. Cir. 2020). This same treatment should extend to dispatchable hybrid resources more generally.

⁵⁹ SPP, Filing, Docket No. ER26-247-000, Transmittal Letter at 26 & Proposed Tariff, Attachment BB, Section 3.1.3(5)(a) (filed Oct. 24, 2025). SPP also proposes to allow a generating facility to support multiple large loads if no more than five substations are involved and no more than two existing transmission line segments are used between each substation. *Id.*, Transmittal Letter at 27 & Proposed Tariff, Attachment BB, Section 3.1.3(5)(a).

- (ii) What curtailment and dispatch commitments may qualify a load or hybrid facility for queue priority?

We recommend that the Commission provide for queue priority to loads and hybrid facilities that have either made “capacity flexibility commitments” or “transmission flexibility commitments.” This matches the need for reform, which stems both from the challenge that rapidly integrating load creates with regard to Network Resource and capacity procurement, and from the fact that current interconnection processes for large loads and hybrid facilities trigger more network upgrades than necessary which raises costs for transmission customers and slows the interconnection process.

The flexibility commitment necessary to avoid transmission network upgrades is different than the commitment necessary to avoid capacity procurement. The former is a commitment to curtail/dispatch in response to potential overloads or exceedances on the transmission system, must be subject to the system operator’s control in real time as such conditions may arise in response to unexpected contingency occurrences, and must be of sufficient duration to warrant temporary or permanent avoidance of network upgrades. The latter is a commitment to curtail/dispatch in response to potential capacity shortfalls, which may be known hours in advance or in the day-ahead time frame. Because these two types of commitments are different, and may have different frequency or duration requirements in different locations, some resources may have the commercial or technical capability to make one type of commitment but not the other. Entities that commit to flexible operations that avoid capacity procurements but cannot commit to flexible operations that avoid network upgrades should be allowed to do one but not the other. But both offer significant system benefits. As such, we recommend that FERC provide that resources that have made either type of commitment be eligible for queue priority in

the load interconnection process. Resources that can make both commitments should be eligible for additional speed-to-power benefits through a streamlined interconnection study process, where they meet the conditions required for that process (see below).

(iii) How will such commitments be enforced?

The mechanism for enforcing a flexibility commitment differs depending on its nature. With regard to transmission flexibility commitments, the relevant curtailment and dispatch commitments are to the transmission provider. These commitments can be enforced through the transmission service provided to the transmission customer that serves the load or hybrid facility. As explained further in subsection c below, to fulfill the need for reform, we recommend that Commission require transmission providers to offer non-firm and conditional-firm transmission service, where the terms of service would provide the transmission provider the right to curtail. In a Notice of Proposed Rulemaking, the Commission can seek comment on whether curtailment and dispatch commitments should additionally be made by load and hybrid facilities directly to the transmission provider.⁶⁰

For capacity flexibility commitments, the commitment to curtail is made to the relevant balancing authority. Accordingly, the Commission should establish requirements for balancing authorities to recognize and enforce commitments to be curtailable/dispatchable in order to reduce resource adequacy needs. Where the transmission provider and the balancing authority are the same entity, the interconnection customer's commitment to curtail/dispatch in response to resource adequacy events could be incorporated into the Interconnection Agreement. However,

⁶⁰ We note that to the extent curtailment and dispatch commitments are made by the transmission customer to the transmission provider, the entity making the commitment could be a load serving entity, which would be distinct from the large load itself. A separate commitment from the large load to the load serving entity may be appropriate in such circumstances, but whether and how to regulate such a commitment would be left to the applicable state regulator or local authority.

as a default, the Commission should provide for a requirement on the balancing authority that is separate from the *pro forma* Interconnection Procedures and Agreement that it may develop through this rulemaking process. The Commission should also promulgate rules to ensure that loads and hybrid facilities that have made Capacity Flexibility Commitments are able to take transmission service without securing commensurate Network Resources, and to ensure that the flexibility commitments of large loads and hybrid resources can be employed to reduce capacity procurements in those regions where capacity is procured through Commission-jurisdictional tariffs.

(iv) What does queue priority mean in practice?

How the transmission provider may prioritize the interconnection requests of loads and hybrid facilities that have made flexibility commitments depends on the overall structure FERC adopts. In a serial study process, FERC could require the transmission provider to grant priority by pushing load and hybrid facilities meeting a particular threshold level of flexibility to the front of the line. FERC could employ a strict requirement, through which such priority is not granted unless the load or hybrid facility can fully curtail or offset its entire load through dispatch of the generation or storage resource portion of the hybrid facility. Alternatively, to the extent the serial process entails a window to submit requests prior to commencing study, the Commission could provide for priority order to be granted based on the extent of the curtailment and/or dispatchability commitment, such that a resource with partial curtailment capability gets priority over a resource with no such capability. Should the Commission opt for a cluster-based load interconnection process, the Commission could require that the transmission provider provide for a cluster formation window that distinguishes between standard load interconnection requests and curtailable and dispatchable hybrid facilities. Under this approach, curtailable and

dispatchable hybrid facilities would be processed first as a priority cluster, before the transmission provider addresses the non-flexible standard load interconnection requests, which would be given lower queue priority.

In providing these illustrative examples of how queue priority could be given to flexible loads and hybrid facilities under established serial and cluster study processes, these comments do not imply an endorsement of these established processes. To the contrary, Eolian supports more ambitious approaches to reform that would use FERC's assertion of jurisdiction over load interconnection as an opportunity to rethink broken paradigms that will only come under increased strain as more large loads seek to interconnect to the system. For example, NRG recommends that the Commission "mandate the use of open seasons to expedite large load interconnections and efficiently allocate interconnection capacity."⁶¹ Such a process could likewise be designed to prioritize curtailable loads and curtailable and dispatchable hybrid facilities. For example, system "headroom" to be auctioned off could be defined in a manner recognizing that curtailable loads and hybrid facilities need less headroom to serve equivalent demand. A system geared toward "getting the most megawatts online at the lowest overall cost" would naturally reward flexible loads and hybrid facilities, so long as it is paired with appropriate study processes and non-firm transmission service (discussed below). Alternatively, the Commission may also consider an approach that integrates transmission planning with load interconnection and generation interconnection, similar to how SPP is proposing to integrate transmission planning and generation interconnection through its Consolidated Planning Process.⁶²

⁶¹ NRG Energy, Inc., Initial Comments at 3.

⁶² See SPP Transmittal Letter, FERC Docket No. ER26-414 (Nov. 3, 2025).

b. Rules requiring a **streamlined study process** to allow flexibility and dispatchability to avoid network upgrades

Load interconnection queue priority, however, must not be the only way in which the Commission expedites the interconnection of curtailable loads, and curtailable and dispatchable hybrid facilities. As demonstrated in Testimony of Mr. Levitt, Dr. Mohan, Dr. Quint, and Ms. Tanneeru, the most powerful way that the ANOPR can accelerate speed-to-power while also protecting existing transmission and wholesale electricity customers is to allow large loads and hybrid resources to use their flexibility to avoid, minimize or defer the need for new network upgrades, which are both a major source of delay and a cost driver. To achieve this objective, the Commission should require each transmission provider to develop a load interconnection study process that incorporates the interconnection customer's commitment to curtail/dispatch in response to specific transmission system needs and allows for that commitment to reduce or eliminate the need for new network upgrades. As a consequence of making this commitment, the large load or hybrid resource would agree to be subject to the system operator's control for those curtailment/dispatch periods, and would be required to demonstrate its technical capability to effectuate the necessary curtailment/dispatch.

We propose establishment of a **serial streamlined study** of load and hybrid facilities as discussed in detail in Section VII, which can fulfill this need. We recommend that the Commission require this proposed study process as part of *pro forma* Hybrid Facility Interconnection Procedures.

c. Non-firm injection and withdrawal rights will be necessary to realize the economic benefits of hybrid resources

The ANOPR presents the example of a hybrid resource consisting of a 600 MW generator and a 500 MW load. The ANOPR states that, in such a scenario, the hybrid resource may elect to receive 100 MW of injection rights and no withdrawal rights, while also noting that “Any facility should be able to seek additional rights through a separate interconnection request.”⁶³ To address this scenario, the Commission should direct transmission providers to offer non-firm or conditional firm transmission service to hybrid resources that allow them to inject and withdraw up to their full potential, subject to specified system conditions and their flexibility commitments.

Without access to non-firm or conditional firm service, much of the economic value of the hybrid resource model will be lost. Many hybrid resources would be unwilling to interconnect with no withdrawal rights at all, because the load would bear the risk of generator outages (especially risky for facilities lacking many smaller, redundant units), which can be lengthy and/or unexpected, and, in the case of data centers, exhaust the ability of backup generation to compensate. Providing hybrid resources access to non-firm or conditional firm withdrawal service would therefore go a long way toward de-risking the hybrid resource business model. Further, without access to non-firm or conditional firm withdrawal service, any standalone storage component of a hybrid resource would be unable to charge, which could have the effect of excluding from the market the most rapid-to-deploy source of capacity for integrating large loads.

⁶³ ANOPR at P 22 & n. 39.

Access to non-firm or conditional firm injection service for the generator component of a hybrid resource would also provide substantial economic benefits. Most obviously, at times when the load component is operating at a low load factor, allowing the generator to supply the broader market on a non-firm basis will benefit other customers on the system while providing an important revenue stream to the hybrid facility that may undergird the investments that enable its flexibility commitments. Further, if the generator component of a hybrid resource is unable to inject its unneeded generation, then it may be effectively forced into a load-following role, which is extremely difficult to sustain indefinitely for individual generators, “even for the most flexible on-site generation, reducing the commercial workability of the hybrid resource model.”⁶⁴

d. Rules requiring provision of generator option to build

In addition to the above rules to recognize the value that flexible loads and dispatchable hybrid resources can offer the grid and facilitate their development, Eolian supports the ANOPR’s recommendation that interconnection customers be afforded the option to build.⁶⁵ The option to build a transmission provider’s interconnection facilities and network upgrades has been available to generator interconnection customers in the *pro forma* LGIA since Order No. 2003, and as later enhanced in Order No. 845. Large load interconnection customers will benefit from the option to build for all the reasons the Commission determined it was just and reasonable to provide this option in the generator interconnection process: it “will benefit the interconnection process by providing interconnection customers more control and certainty during the design and construction phases of the interconnection process.”⁶⁶ The option enables

⁶⁴ See Eric G. Gimon, *Dodging The Firm Fixation For Data Centers And The Grid*, ENERGY INNOVATION, at 12 (Nov. 2025), <https://energyinnovation.org/wp-content/uploads/Dodging-the-Firm-Fixation-for-Data-Centers-and-the-Grid.pdf>.

⁶⁵ ANOPR at P 26.

⁶⁶ *Reform of Generator Interconnection Procs. & Agreements*, Order No. 845, 163 FERC ¶ 61,043, at P 85 (2018), *order on reh’g*, Order No. 845-A, 166 FERC ¶ 61,137, *order on reh’g*, Order No. 845-B, 168 FERC ¶ 61,092 (2019).

an interconnection customer to potentially build the facilities in a more cost-effective manner, leading to lower cost interconnection.

The Commission should build upon its experience in the generator interconnection process by further reducing barriers to use of the option to build. Currently rules for the generator option to build are heterogenous and in practice this option is hard to access and highly disincentivized, despite the ability to save considerable money, time, or both. A greater degree of standardization of the option will eliminate opportunities for undue discrimination in access, and better realize its significant benefits. For example, Eolian has experienced receiving a quote for a specific project from an Engineering, Procurement, and Construction developer and then being quoted significantly more for the utility to perform the same project. In addition, developer option to build is effectively limited to only those upgrades which require no outage coordination, sharply limiting the scope of its availability.

Given the ANOPR's emphasis on speed and efficiency, we encourage FERC to consider rules that would bring greater consistency and transparency to this option by requiring cost reporting and auditability of costs and the ability for developers to bid out the same scope—paying more for speed if desired.

VII. STREAMLINED, SINGLE STUDY, SERIAL INTERCONNECTION PROCESS FOR HYBRID FACILITIES

So long as transmission system capacity remains constrained, the rate-limiting step for integration of large loads will remain the many years-long process of constructing network upgrades.⁶⁷ Certain configurations of hybrid facilities, however, can make such significant

⁶⁷ Although, as noted above, in regions that require transmission customers to demonstrate that a new load will not threaten resource adequacy or that it has secured commensurate Network Resources before being permitted to interconnect, interconnection of new generation will also increasingly rate-limit the interconnection of large loads. *See supra* n.47.

flexibility commitments as to enable greatly reduced study times, and are capable of being developed much more rapidly due to their limited impacts on the transmission system. Where such configurations are commercially feasible, the Commission should incent these arrangements by providing them a separate streamlined, single study, serial interconnection process. Such a **streamlined study and interconnection process** would enable, to a greater degree than any other reform, near-term, rapid development and integration of eligible hybrid facilities.

As set forth in Attachment A, serial streamlined study of load and hybrid facilities is possible where the facility nominates operating limits (in MW) for aggregate net injections and withdrawals of power,⁶⁸ installs necessary control equipment to ensure that the facility is capable of following the dispatch controls of the transmission system operator and balancing authority and respecting operating limits at all times, and is capable at any time of complete self-supply or otherwise curtailing the load such that the facility can take transmission service on a non-firm basis.⁶⁹ Where these conditions are met, the facility can be studied without carrying out generator and load deliverability analysis.⁷⁰ As Levitt et al. explain, “[b]ecause the streamlined process avoids major network upgrades (if necessary, by limiting both net injections and net withdrawals as low as zero megawatts), each transmission owner can independently conduct streamlined hybrid interconnection processes (even within an ISO/RTO region) in serial fashion in as little as 60-90 days.”⁷¹ Importantly, these studies can be led by individual transmission providers within an RTO. In addition, “avoiding regional upgrades reduces interactions among

⁶⁸ For large loads that are not hybrid, the operating limits would entail withdrawals only.

⁶⁹ See Attachment A at 14 (describing operational obligations) and 14-17 (describing studies and mitigations, as well as processing of studies).

⁷⁰ Studies for such a facility include (i) steady-state power flow (thermal and voltage) analysis, (ii) stability analysis, (iii) short circuit and fault duty analysis, and (iv) electromagnetic transient (EMT) stability screening and detailed EMT impact studies where needed). See *id.* at 15.

⁷¹ *Id.* at 10.

transmission owners, allowing each transmission owner to conduct streamlined hybrid facilities studies independently . . . which increases the number of such studies that can be undertaken simultaneously in a region.”⁷² Accordingly, “[a]n ISO/RTO with two dozen transmission owners could therefore process up to 96 sets of hybrid facility interconnection requests over the course of a year, with four 90-day study cycles a year and 24 of them running in parallel at any one time.”⁷³ FERC should require transmission providers to offer a serial streamlined study and interconnection process to loads and hybrid facilities that meet these requirements.

The streamlined study process specifies the static operating limits below which the load or hybrid facility can be integrated into the system without triggering network upgrades.⁷⁴ However, developers may desire network upgrades in order to obtain a higher operating limit and/or reduce expected curtailment. FERC should require that in such cases, the facility first have the option to avoid the critical impact through lower operating limits.⁷⁵ If the developer wishes to raise the operating limit, however, the facility should be transferred into the applicable standard generation or load interconnection process. To the extent a potential need for network upgrades is triggered by the injection rights requested by a hybrid facility, FERC should require that the facility be studied through the applicable generation interconnection process. To the extent the need is triggered by the facility’s withdrawal rights, the impacts can be studied through the standard load interconnection process. In such cases, it is still possible to study the facility’s generation components in a serial, streamlined fashion outside of the standard generation interconnection queue, but energization of the generation component can be conditional on completion of the applicable load interconnection process. And in each case,

⁷² *Id.* at 16.

⁷³ *Id.* at 10 (internal citation omitted).

⁷⁴ *Id.* at 12-13.

⁷⁵ *Id.* at 14-16.

where the load and electrically close supply components of the hybrid facility are commercially dependent upon one another, the load interconnection process should account for the associated generation, and the generation interconnection process should include the load. Accordingly, transmission providers should be required to amend their large generator interconnection procedures and associated agreements to provide for such a streamlined study process where generation at the same or electrically close point of interconnection to a large load can be studied without requiring deliverability analysis.

Flexibility commitments can be made by loads and hybrid facilities on a permanent or temporary basis (i.e., until network upgrades have been constructed), and FERC should require transmission providers to allow for both pathways. Temporary commitments are valuable because they provide the ability of the load or hybrid facility to integrate into the system while potential network upgrades are analyzed and constructed. Temporary commitments can be facilitated by allowing a load or hybrid facility to convert from non-firm to firm service by requesting a full load interconnection study through the standard load interconnection study processes, and/or to request study of the generation components through the standard generation interconnection process.⁷⁶ Where network upgrades are identified as necessary in order to facilitate a conversion in service, the facility's service conversion may take place once the network upgrades are constructed.

Importantly, however, FERC should also require that transmission providers allow a pathway to permanent non-firm operation that does not require a subsequent service conversion. Requiring subsequent service conversion greatly diminishes the commercial value of pursuing flexibility arrangements. For example, a hybrid facility consisting of a co-located battery or gas-

⁷⁶ *Id.* at 4.

fired power plant requires the developer to finance these assets, which have a longer-asset life than the period of time that would be required to build network upgrades. Where a load or hybrid facility has equipment that permanently enables a network upgrade to be avoided, requiring a service conversion will needlessly raise costs for customers. Further, because any ultimate conversion to firm service would be potentially contingent upon network upgrades whose costs are unknown, the developer may be unwilling to risk investing capital in building the equipment required to facilitate the temporary service arrangement.

VIII. CONCLUSION

This rulemaking proceeding presents an enormous and critical opportunity to address the significant challenges facing our industry. Seizing this opportunity to protect consumers without constraining the economic development potential that large load growth represents will be complex, requiring revisions to different sets of tariff provisions that have traditionally been subject to separate regulatory requirements and Commission precedent, including load interconnection processes, generator interconnection processes, transmission service requirements, and Commission-jurisdictional resource adequacy constructs. Getting these reforms right must be the first priority; the Commission should take the time needed to ensure that any final rule issued adequately addresses the full scope of problems driven by large load growth in a systematic way. Specifically, for the purposes of this comment, failure to truly enable flexible loads and provide them a means to internalize the benefits they provide to the system risks *slowing down* progress by disrupting the potential for state innovation and failing to reduce the monumental cost of transmission expansion and capacity growth otherwise necessary to integrate large loads into the transmission system. Eolian is optimistic, however, about the Commission's ability to issue such a rule, and greatly appreciates the opportunity to provide a

workable framework and set of pathways that will enable flexibility resources, and, in particular, hybrid resources to contribute to solving these problems to the maximal extent technically feasible.

Respectfully submitted,

/s/Ariel Horowitz

Ariel Horowitz

Eolian L.P.

988 Howard Ave. Suite 200,

Burlingame, CA 94010

ariel.horowitz@eolianenergy.com

Miles Farmer

Avi Zevin

Nathan Lobel

Roselle LLP

300 New Jersey Ave. NW, Suite 300

Washington, DC 20001

miles.farmer@rosellellp.com

avi.zevin@rosellellp.com

nathan.lobel@rosellellp.com

Counsel for Eolian L.P.

Attachment A

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

**Interconnection of Large Loads to the
Interstate Transmission System**

Docket No. RM26-4-000

**AFFIDAVIT OF
ANDREW LEVITT, ANIRUDDH MOHAN, RYAN QUINT, AND SIRISHA TANNEERU
ON BEHALF OF
EOLIAN ENERGY, INC.**

November 21, 2025

I. QUALIFICATIONS

Our names are Andrew Levitt, Aniruddh Mohan, Ryan Quint, and Sirisha Tanneeru. Mr. Levitt is a Senior Consultant and Dr. Mohan is an Energy Associate at The Brattle Group (Brattle). Dr. Quint is the President and Chief Executive Officer and Ms. Tanneeru is Lead Engineer at Elevate Energy Consulting (Elevate).

Mr. Levitt is an expert on wholesale market design and reliability implications of changes in the power system. He has consulted to Independent System Operators and Regional Transmission Organizations (ISO/RTOs) such as the Midcontinent Independent System Operator (MISO), the New York Independent System Operator (NYISO), the Alberta Electric System Operator (AESO), and the Independent Electricity System Operator (IESO) of Ontario, and helped other clients navigate ISO/RTO changes nationwide. Prior to joining Brattle in 2022, Mr. Levitt was on staff at PJM for seven years (most recently as Lead Market Designer), where he developed reforms to reliably integrate new resource types. Mr. Levitt holds an MMP from University of Delaware and a BSc in physics from University of Toronto.

Dr. Mohan specializes in the evaluation of emerging energy technologies at a systems level, with expertise in electricity markets, techno-economic analysis, and energy policy. He has expertise in optimization models that support both resource and transmission planning and has published peer reviewed research in leading academic journals on the integration of emerging technologies in energy systems. He holds a PhD in Engineering & Public Policy from Carnegie Mellon University.

Dr. Quint has participated in and led numerous industry initiatives at the Energy Systems Integration Group (ESIG), Institute of Electrical and Electronics Engineers (IEEE), International Council on Large Electric Systems (CIGRE), and North American Electric Reliability Corporation (NERC) forums. Prior to founding Elevate, Dr. Quint spent nearly 9 years at the NERC, most recently as the Director of Engineering and Security Integration where he led strategic initiatives working with industry stakeholders focused on emerging reliability risk mitigation. Dr. Quint holds a PhD in power systems engineering from Virginia Polytechnic Institute and State University and is a registered Professional Engineer.

1 Sirisha Tanneeru has over 15 years of experience in the electric utility industry, spanning across
2 utilities, renewable development, and consulting. She has extensive experience in transmission
3 expansion planning, particularly generation and load interconnections and NERC compliance.
4 She provided interconnection strategy support for several greenfield and brownfield renewable
5 projects in the MISO, Southwest Power Pool (SPP), and the Western Electricity Coordinating
6 Council (WECC) regions. Ms. Tanneeru holds an MS from New Mexico State University.

7 **II. ASSIGNMENT**

8 We were asked by Eolian Energy, Inc. to comment on the Secretary of Energy's Advance Notice
9 of Proposed Rulemaking (ANOPR), specifically to substantiate and assess reliability aspects of
10 the fourteen principles associated with studies and operations of hybrid facilities (i.e., generation
11 facilities supporting on-site or nearby large load facilities) that can meet their own capacity needs
12 without relying on the broader system for firm power. We focus on ANOPR principles numbered
13 three, five, six, and seven.

14 **III. SUMMARY OF PROPOSED PROCESS TO STREAMLINE CONNECTION OF** 15 **HYBRID FACILITIES**

16 The ANOPR proposes reforms to the process for interconnecting large loads to the transmission
17 system guided by fourteen principles, several of which bear on studies and operations of hybrid
18 facilities. The Federal Energy Regulatory Commission (FERC) has initiated a rulemaking to
19 consider and take final action on the ANOPR, which may result in a NOPR and a subsequent
20 final rule.

21 In this Affidavit, we develop and assess a proposal for one aspect of the streamlined
22 interconnection study process specified in principle seven, while also addressing principles three,
23 five, and six. **Our proposal relies on the key capability of a hybrid facility to meet its own**
24 **power needs—i.e., achieve a zero net megawatt (MW) exchange with the grid—when so**
25 **dispatched at any time (including load curtailment if the hybrid's facility's supporting**
26 **generation cannot produce).** The potential for "net zero impact" for a hybrid facility, including
27 the ability of the load to be curtailed if the generation is unavailable, is a compelling case to

1 simplify the study process, minimize costly network upgrades, and maximize system operator
2 controllability to ensure reliable operation of the bulk power system._

3 Because new supporting generation that is electrically close to a new large load (but not
4 necessarily on-site) can provide the same benefits as a hybrid facility, our analyses include this
5 broader configuration (following SPP's recent High Impact Large Load Generation Assessment
6 (HILLGA) proposal that similarly allows for loads and generation that are electrically near each
7 other but not necessarily at the same site).¹ We refer to both on-site and electrically close as
8 hybrid facilities. We recommend that the Commission's proposed definition of hybrid facilities
9 include both on-site and electrically close designs of hybrid facilities.

10 This proposal sets forth a streamlined study process for connecting a new hybrid facility. As
11 explained further below, where such a facility meets certain flexibility requirements, its
12 interconnection to the system can be streamlined by its interconnecting Transmission Owner in a
13 serial fashion outside of the standard interconnection queues for either generation or load. In
14 such circumstances, only one study process is required to connect both the load and generation
15 components of the facility. The streamlined process yields transmission agreements and control
16 mechanisms for connection and service for both the generation and load components of the
17 hybrid facility, granting the necessary conditional permission for injection and withdrawal.
18 Should the serial study process we set forth find major transmission network upgrades are likely
19 to be needed at the requested level of service, then it identifies a lower operating limit at which
20 the hybrid facility can connect without triggering such upgrades. In order to increase the
21 operating limit or to request conventional firm service, the hybrid facility developer has the
22 option to request a full load interconnection study through the standard load interconnection
23 study process, and/or to require the generation components of the hybrid facility to be studied in
24 the standard generation interconnection process (but in a manner that accounts for the existence
25 of the associated load).

26 While the proposal is focused on dispatchable and curtailable hybrid facilities, the streamlined
27 study methods we set forth are inherently applicable to curtailable large loads as well. Like a

¹ Southwest Power Pool, [Submission of Tariff Revisions to Add the High Impact Large Load Processes and Generation Assessment](#), filed October 24, 2025, FERC docket number ER26-247

1 hybrid facility, a curtailable large load could also avail itself of the standard load interconnection
2 study process in order to increase the operating limit identified in the streamlined process, or to
3 firm up its service to avoid curtailment.

4 The streamlined study process we set forth is one mechanism through which the Commission can
5 provide for quicker interconnection of curtailable and dispatchable hybrid facilities. As we
6 explain, this study process, which allows for serial study of facilities by the interconnecting
7 transmission owner outside the queue, is only appropriate where the facilities can meet a strict
8 set of operational and other criteria.

9 The proposed approach to rapid interconnection of hybrid facilities, which is in many respects
10 similar to SPP's recent HILLGA proposal, has the following features:

- 11 • **Is led by the transmission-owning utility**, following the current load
12 interconnection processes.
- 13 • **Jointly studies** load and supporting generation facilities.
- 14 • **Offers rapid study to confirm minimal material impact**, conceptually
15 similar to (although with a more thorough study than) the streamlined
16 processes currently used for Surplus Interconnection Service in MISO and
17 SPP, which quickly identifies only *de minimis* network upgrades.²
- 18 • **Utilizes the same studies already used today, except without deliverability**
19 **studies** for firm net injections or withdrawals, since the hybrid facility relies on
20 self-supply for firmness rather than on the network (and the hybrid facility can
21 limit its net grid injections or withdrawals whenever necessary).
- 22 • **Resolves study violations through operational means (rather than physical**
23 **upgrades) whenever practical and reliable**; for example, steady state pre-
24 contingency violations that can be addressed through operational curtailment
25 via Security Constrained Economic Dispatch (SCED) do not require network
26 upgrades.

² GridLab, [Surplus Interconnection: A Tool to Immediately Deploy New Electricity Supply to Enhance Grid Reliability and Affordability](#), January 2025

- **Relies on static operational limits** where necessary to avoid transmission upgrades by limiting the maximum allowable net grid injections or withdrawals to certain study-determined levels (or even zero) at any time.

We have assessed the timing of studies necessary to implement the proposed approach, and conclude that, in cases in which the process does not flag the need for additional detailed analysis, studies can be completed in 90 days—and potentially as quickly as in 60 days.

Finally, we observe that a hybrid facility that can self-supply its energy needs at any time (or otherwise curtail the load) can also use that capability to self-supply its resource adequacy requirements so that no resource adequacy responsibilities are imposed on the rest of the system. A Balancing Authority can recognize this in its resource adequacy construct (e.g., by not allocating capacity obligation to the load of such a hybrid facility and by ensuring that the dispatch and curtailment obligations of the hybrid facilities apply to both resource adequacy and transmission reliability events).

III.A. Benefits of Rapid Interconnection of Hybrid Facilities

The proposed reforms to rapidly study and connect hybrid facilities take advantage of the fact that their proximity and dispatch flexibility limits impacts on the broader transmission system at any time. This provides the below benefits.

Accelerates Economic Development Priorities: the proposed reforms apply reliability analysis techniques to closely coordinate transmission planning studies with state-of-the-art grid operations, enabling faster connection of data centers and other large loads that fulfill national and state priorities for artificial intelligence, critical energy infrastructure investments, electronics, critical minerals, and other public priorities.

Encourages Entry of New Supply: by making the avoided transmission value of new supply available to adjacent large load customers in the form of accelerated hybrid interconnection, the proposed reforms provide a substantial incentive for new loads to contract with and bring online new supply. Meanwhile, joint studies take advantage of adjacency to bring new supply on more

1 quickly than would otherwise be possible. Compared to large load growth without such reforms,
2 this aspect of the proposed streamlined hybrid study process offers these benefits:

- 3 • **Mitigation of Resource Adequacy Concerns:** the timely addition of on-site or
4 electrically close new supply that would not otherwise enter service (or only
5 with significant delays) mitigates both transmission capacity and resource
6 adequacy concerns.
- 7 • **Reduction in Capacity Prices:** the additional supply relieves scarcity in
8 markets for capacity and resource adequacy contracts, supporting affordability
9 objectives for existing customers, including in restructured states.³
- 10 • **Reduction in Energy Prices:** greater supply (especially during tight periods)
11 reduces average wholesale electricity prices.

12 **Facilitates Practical Large Load Flexibility:** by structuring hybrid facility operations such that
13 the load component relies on the dispatch of the associated generation component, utilizing load
14 curtailments only as a backstop measure, the proposed reforms unlock load flexibility in a way
15 that is compatible with large load business models that rely on very high availability—that is,
16 outsourcing much of the necessary overall flexibility to a supporting on-site or nearby generation
17 facility. This allows large loads to continue to count on exceptionally high availability of nearby
18 generation, while leveraging load curtailments (or backup generation) during those rare
19 occasions when the hybrid facility’s nearby generation resource cannot produce.

20 **Limits Impact on Conventional Interconnection Requests:** because the benefits of the reforms
21 are achieved by recognizing the capability of hybrid facilities to limit net grid injections and
22 withdrawals and their impacts on the transmission system (rather than by re-assigning available
23 grid capability to them from other generators with interconnection requests), the proposed
24 approach has less impact on conventional generator and load interconnection requests.

25 **Can Rely on Protection Systems to Avoid Unauthorized Grid Use:** the proposed reforms
26 require that inadvertent net injections or withdrawals of electricity from the transmission system

³ For example, an analysis by the Independent Market Monitor for PJM found that rising capacity costs were almost entirely due to the resource adequacy needs of new large loads seeking to connect to the grid.

1 in excess of standing operating limits can be avoided with site-level or local protection schemes,
2 which may be either behind-the-meter or local contained protections. This requirement simplifies
3 reliability study methodologies and the treatment of the hybrid facility's component outages.
4 These protections do not need to be classified as "remedial action schemes," particularly when
5 they are customer-controlled/owned.⁴

6 **IV. PROPOSED PROCESS TO CONNECT HYBRID FACILITIES**

7 To support development of a proposed rule for rapid interconnection of hybrid facilities, we
8 describe high-level features of the proposed process.

9 In addition to generally providing direction to FERC to regulate the interconnection of new loads
10 greater than 20 MW to the transmission system, the ANOPR distinguishes between the
11 interconnection of firm loads and hybrid facilities and those that are curtailable and dispatchable.
12 This most obviously covers two use cases important for accelerating interconnection, addressing
13 transmission needs, and mitigating resource adequacy shortfalls: (1) curtailable large loads and
14 (2) co-located hybrid facilities that can reduce net injection/withdrawals on demand. The
15 ANOPR's concept of hybrid facilities can be extended to cases in which the large load is
16 electrically close to the paired generation, metered separately,⁵ and not necessarily at the same
17 site. This "electrically close" hybrid approach can provide many of the same benefits as strictly
18 on-site generation and load but is likely to be more commercially feasible for some loads and
19 generation and therefore would deliver more value to the ANOPR's objectives than a narrower
20 approach. Such electrically close hybrid facilities allow more flexibility in terms of their
21 development and siting, and do not require complex contracting to enable the on-site self-supply
22 of power that may be inconsistent with state laws on utility franchises (or direct retail sales
23 where applicable). Studying electrically close load and generation as a single hybrid facility adds
24 some complexity relative to studying a hybrid facility with on-site (and behind the meter) co-
25 located loads and generation (for example to address the potential for localized overloads);

⁴ NERC governs remedial action schemes through standard [PRC-012-2 - Remedial Action Schemes](#).

⁵ Large loads that are contracted with supporting generation within the same resource adequacy zone but are not electrically close such that they cannot offer the same transmission cost and speed benefits as on-site or electrically close hybrid facilities, fall outside our proposed definition—even though we recognize that they can still meet the resource adequacy needs of a balancing authority.

1 however, we find that this complexity is unlikely to add significant workload or slow down the
2 study process.

3 Because flexible loads, on-site hybrid facilities, and electrically close hybrid facilities can all
4 address both resource adequacy needs and transmission reliability needs, they should all be
5 enabled by the ANOPR. A streamlined interconnection option can be made available to all of
6 these options whenever the operating arrangement avoids time-consuming grid upgrades. These
7 fast-track options can presume that majority of the grid impacts are managed in the operations
8 horizon through SCED (including pre-contingency dispatch to ensure N-1 security) via dispatch
9 of generation (backstopped by curtailment of load), in combination with standing operational net
10 injection and withdrawal limits established in the streamlined interconnection studies.

11 Allowing for non-firm net injections from a hybrid facility (when the grid can accommodate it)
12 can benefit the wholesale energy market and also support the system during operational resource
13 adequacy shortfalls. Such grid injections would be allowed only on an “as available” basis,
14 subject to system operator (and market-based SCED) dispatch control. This is similar to the
15 Electric Reliability Council of Texas’s (ERCOT’s) “connect and manage” approach to generator
16 interconnection, and even less firm than existing Energy Resource Interconnection Service
17 (ERIS).⁶ Such conditional non-firm net injections based on available grid capacity are broadly
18 similar to PJM’s recent proposal for “provisional service” enabled by “Interim Deliverability
19 Studies”, as well as the Provisional Interconnection Service required under FERC Order Number
20 2023.⁷ Our proposed fast-track process for hybrid facilities is also similar to SPP’s recently
21 proposed 90-day HILLGA process filed with FERC in October.

22 When the streamlined studies detect major planning impacts that cannot be resolved through
23 operational redispatch (and therefore would require network upgrades), the process identifies the

⁶ This level of very non-firm net injection service is well matched to generators whose primary purpose is to serve a local load, rather than to make net injections to the grid. However, developers of non-hybrid generators may also be interested in this level of service. While beyond the scope of this affidavit, we note that similar benefits to the energy market would be possible if the service were made available to all generators, and that such service could likely be processed more quickly than conventional ERIS or Network Resource Interconnection Service (NRIS), as evidenced by the comparative speed advantage in ERCOT. See e.g. LBNL, [Queued Up: 2024 Edition](#), April 2024, slide 35.

⁷ D. Bielak, [PJM Manual 14: New Service Requests Cycle Process v3 Revisionary Summary](#), 2025; FERC, [Order No. 2023](#), issued July 28, 2023, 184 FERC ¶ 61,054

1 lower operating limit that can be accommodated without such network upgrades. The hybrid
2 facility's developer then has the option to connect with the identified lower operating limits (on
3 either net injections or net withdrawals or both), and/or to proceed with additional studies
4 through the usual interconnection processes (i.e., the generator interconnection or the
5 conventional load integration processes, or both) in order to pursue the network upgrades needed
6 to increase the applicable operating limit (and/or to decrease expected curtailment risk).⁸ For
7 example, if a hybrid facility requesting to inject and withdraw certain amounts were flagged for
8 major grid upgrades to support both, it could choose to immediately connect with the lower net
9 injection limit and the lower net withdrawal limit (identified by the transmission owner,
10 potentially as low as zero megawatts), while simultaneously pursuing conventional load and
11 generator interconnection request to increase one or both operating limits in the longer-term. The
12 combination of streamlined studies allowing for immediate connection plus standard
13 interconnection process to enhanced operating limits and firmness allows for an efficient and
14 coordinated network upgrade process among all affected transmission systems.

15 Because the streamlined process avoids major network upgrades (if necessary, by limiting both
16 net injections and net withdrawals as low as zero megawatts), each transmission owner can
17 independently conduct streamlined hybrid interconnection processes (even within an ISO/RTO
18 region) in serial fashion in as little as 60–90 days. Meanwhile, any network upgrades desired by
19 the customer to increase the operating limit on net withdrawals or net injections can be processed
20 through the conventional interconnection process (on a slower timeline). An ISO/RTO with two
21 dozen transmission owners could therefore process up to 96 sets of hybrid facility⁹
22 interconnection requests over the course of a year, with four 90-day study cycles a year and 24 of
23 them running in parallel at any one time, even if each transmission owner can only conduct one
24 at a time. However, in some cases a transmission owner may be able to study certain hybrid

⁸ We note that the generator interconnection process and the load integration process should both be enhanced with the study principles described here, namely: (a) joint study of contractually linked and electrically close generation and load; (b) recognition of curtailment as a reliability solution, if elected by the requestor and achievable in operations; and (c) recognition of fixed operating limits as a method to avoid upgrades where desired.

⁹ Coordination of impacts among transmission owner regions is required when hybrid requests are electrically close to each other. Such coordination can be established upfront such that the study includes all impacts without requiring separate affected system transmission owner analysis.

1 facility requests in proximity together, so that across the ISO/RTO they can potentially process
2 more than the 96 requests per year.

3 IV.A. Applicability and Qualification

4 To illustrate the proposed fast-track process for hybrid facility interconnection studies, we focus
5 this discussion on the case of an agreement between a large new load and a new generator that
6 are electrically close but not at the same site. While our discussion focuses on such electrically
7 close hybrid facilities, the process is equally applicable to large loads that are curtailable and to
8 hybrid facilities with on-site loads and generation, such as those configured as energy parks with
9 behind-the-meter generation that can control their grid withdrawals and injections as required.
10 Under this process, hybrid facilities qualify for streamlined interconnection when they can
11 maintain specified operational limits that will be identified in the interconnection study, meet
12 dispatchability requirements, and meet other applicable requirements such as readiness
13 requirements, deposits, and a completed application.

14 The streamlined study framework would apply to a hybrid facility's generation resource even if
15 the facility requests firm withdrawal rights for the entire load. While doing so would require the
16 load interconnection request to be studied through the standard load interconnection process, the
17 streamlined study process could still be applied to the hybrid facility's generation
18 interconnection request as grid net injections from the hybrid facility would still be controllable
19 (e.g., to zero MW). Energization of the generation component can be conditional on completion
20 of the applicable load interconnection process. A generator and load of a hybrid facility are
21 "electrically close" when power flows directly from the former to the latter within the confines
22 of the local area. This "local area" is the set of nearby substations to which the hybrid facility's
23 load and generation are connected, and the transmission facilities connecting them. When such
24 self-supply of large loads induces material loop flows over transmission facilities outside this

1 local area, it is not “electrically close”¹⁰ and the generation and load cannot be treated as a hybrid
2 facility for the purpose of the proposed streamlined interconnection process.

3 IV.B. Static Operating Limits and Configurations

4 Static operating limits play an important role in the present proposal. The operating limits
5 represent the maximum curtailable levels of net injections and withdrawal under which the grid
6 can be safely operated without material upgrades. The operating limits are static on a day-to-day
7 basis, with dynamic dispatch of the hybrid facilities taking place within the outer envelope of the
8 operating limits. Therefore, they do not represent firm access to transmission capacity, but rather
9 the limit of non-firm access to transmission capacity.

10 As part of its application for streamlined interconnection, the hybrid facility would nominate
11 initial static operating limits (in megawatts) for aggregate net injections and net withdrawals of
12 power (consistent with ANOPR principle five). The streamlined process studies the nominated
13 operating limits, and if necessary, establishes lower operating limits to avoid network upgrades.

14 The static operating limits are applied to any withdrawals (e.g., the sum of load and any battery
15 storage charging, if applicable) net of any generation output, and vice versa (i.e., injections net of
16 consumption). The grid operator can dispatch the combined load and generation components of
17 the hybrid facility anywhere within the allowed operating limits as needed to ensure reliable and
18 efficient operations, including effective curtailment of the joint withdrawals or injections. When
19 the grid operator dispatches a curtailment below the operating limit, the hybrid facility must
20 reduce power accordingly.

21 To secure reliability, interconnection studies may specify automatic controls to maintain net
22 output within operating limits between dispatch intervals, for example in case of sudden loss of a
23 generation unit within the hybrid. Such automatic controls may be required to ensure the
24 operating limits are respected in a sufficiently timely fashion. When that timescale is less than

¹⁰ An initial concept for electrical closeness could allow for only 3% of self-supply power to flow through ties connecting the local area to the network, and 1% to flow over other transmission facilities outside the local area. See Appendix I, Sirisha Tanneeru, et al, “Reliability Assessment of Hybrid Facilities: Overview of Large Load and Generation Co-located Studies”, slide 5.

1 SCED dispatch frequency (e.g., five minutes in most ISO/RTO markets), it may require the
2 installation of automatic controls to limit net injection/withdrawal from the hybrid facility,
3 potentially instantaneously.

4 As discussed further in Section IV.G below, we propose that static operating limits on net
5 withdrawals are fixed. For the static operating limits on net injections, two alternative policies
6 may be suitable: either operating limits that are perpetually fixed, or operating limits are adjusted
7 periodically (e.g., annually or quarterly), including potential reductions. This could be similar to
8 and coordinated with any existing Provisional Interconnection Service protocols.

9 Various configurations of the hybrid facility are possible, with implications for the
10 corresponding operating limits:

- 11 • **Baseload self-supply:** the hybrid facility's generation matches demand at most
12 times, except for possibly brief periods to allow for rebalancing following a
13 contingency for loss of a hybrid generator unit or portion of the load;¹¹ in this
14 case the operating limits would be zero net withdrawals and zero net injections.
- 15 • **As-available withdrawals:** the hybrid facility's generation matches demand
16 when prices are high or when needed for the grid (including via load curtailment
17 if necessary), but generation never exceeds demand; otherwise, the load is
18 served from the grid as-available; operating limits are net withdrawals for the
19 load, and zero net injections.
- 20 • **Storage-backed hybrid:** same as above (zero net injections) with the option to
21 increase the withdrawal operating limit to accommodate simultaneous storage
22 charging and withdrawal by the large load.
- 23 • **As-available withdrawals and net injections:** when transmission capability is
24 available, the hybrid facility can withdraw from the grid when prices are cheap
25 and inject when prices are high; when needed for the grid, the hybrid facility can
26 self-supply (including load curtailment if needed).

¹¹ Note that these imbalances can also be handled with protection schemes, as mentioned in the ANOPR and above.

IV.C. Operational Obligations

The hybrid facility would have the following obligations:

- Must be available to dispatch to net-zero withdrawal or injection at any time (through increase in generation or curtailment of load, or a combination).
- Must follow dispatch signals for transmission security to reduce net withdrawals or injections, potentially with a higher degree of obligation and control than that faced by an ordinary generator (recognizing that loads may be more reluctant to curtail in operations than an ordinary generator would be to produce power).
- Must maintain net withdrawal and injection to within established static operating limits, including after loss of a portion of the hybrid facility, which may require protection schemes or rapid redispatch depending on the type of transmission security violation.
- When integrated into a resource adequacy construct, must follow dispatches for scarcity mitigation to reduce net withdrawals, with penalties for failure to perform that are comparable to ordinary generators.

IV.D. Studies and Mitigations

The proposed study approach has the following features:

- Studies the generation and load facilities of the hybrid facility in a combined fashion (consistent with ANOPR principle three).
- Studies the combinations of injection and withdrawal necessary to meet the nominated operating limits; if the nominated operating limit cannot be met without triggering network upgrades, the studies identify the lower maximum permissible operating limit (if necessary zero net injections or withdrawals) that can be accommodated;¹² (impacts that would trigger major network upgrades are those that cannot be managed in operations—

¹² Short circuit fault current impacts are completely mitigated through local upgrades

1 this may include stability-based operating limits not captured with SCED or other real-
2 time operational tools).

- 3 • The interconnection studies recognize and incorporate any redundancies in the design of
4 the hybrid facilities and local area that enhance the self-supply capability of the hybrid
5 facility by limiting the impact of loss of any one element; examples include use of many
6 smaller generating units, and substation configurations that avoid the loss of multiple
7 generation units due to failure of a single element.

8 The streamlined hybrid facility connection process conducts the same studies as conventional
9 load and generation connection processes, except that it omits studies of firm deliverability of net
10 injections and withdrawals to and from the broader system (recognizing that firm deliverability
11 instead takes place internal to the hybrid facility's local area). Therefore, the streamlined study
12 process includes:

- 13 • Steady-state power flow (thermal and voltage) analysis, potentially under scenarios such
14 as on-peak and off-peak (and otherwise as ordinarily applicable);
- 15 • Stability analysis;
- 16 • Short circuit and fault duty analysis; and
- 17 • Electromagnetic transient (EMT) stability screening (with detailed EMT impact studies
18 where needed).

19 Unlike the conventional connection process where firm deliverability is required, study
20 violations that can be resolved through operational redispatch and curtailment would not trigger
21 grid upgrades. For example, if studies for a request for 500 MW as-available withdrawals for a
22 hybrid facility indicate that the grid can only accommodate 200 MW securely (and there are not
23 stability-based issues), then no network upgrades are required since system operators can resolve
24 any violations in real-time through curtailment. This is similar to the ERCOT's connect and
25 manage approach to generator interconnection.

26 If stability-based static operating limits are identified, they must be adhered to by the hybrid
27 facility (or network upgrades must be put in place to address the instability if it cannot be

1 detected and managed during real-time operations). Similarly, all short circuit fault duty analysis
2 network upgrades are put in place since they cannot be detected and managed during real-time
3 operations.

4 Since the system must not be put into an unstudied operating state, it is important that the study
5 approach reflects consistency between planning/interconnection assumptions and operational
6 capabilities. This approach is made possible by the operational obligation to be
7 dispatchable/curtailable into a full net-zero withdrawal and injection state at any time,
8 backstopping generation (or exhausted storage) with load curtailment (and/or by dispatching the
9 load's run-time-limited backup generators) if necessary.

10 IV.E. Processing of Studies

11 Because the proposed approach utilizes load curtailment and backup generation as flexibility to
12 manage system impacts, major reliability impacts (that would ordinarily require network
13 upgrades) would only be identified when the issue cannot be managed by operations in real-time,
14 for example via pre-contingency SCED actions. Further, because major network upgrades are not
15 identified in this streamlined approach (upgrades are limited only to the local area), studies can
16 be conducted in a fast-tracked serial study process. Similarly, avoiding regional upgrades reduces
17 interactions among transmission owners, allowing each transmission owner to conduct
18 streamlined hybrid facilities studies independently (similar to load connection processes today),
19 which increases the number of such studies that can be undertaken simultaneously in a region.

20 When the streamlined studies identify critical impacts that would trigger transmission upgrades,
21 a lower operating limit is identified that avoids these critical impacts (potentially as low as
22 zero megawatts). Hybrid facility developers that wish to raise the operating limit have the
23 opportunity to pursue the necessary network upgrades following the standard interconnection
24 study processes and timeline. When a request is to increase the operating limit for net injections,
25 that process is the generator interconnection queue (generally a cluster study process). When the
26 requested operating limit increase is for net withdrawals, the existing load connection process is
27 followed (which can be serial or cluster, depending on the region). Even if request for increased
28 net withdrawals requires that the load interconnection request is studied through the conventional

1 load interconnection process, the generation interconnection portion of the hybrid facility can
2 continue to be analyzed in a streamlined fashion (since net injections would still be limited).
3 Energization of the generation component can be conditional on completion of the applicable
4 load interconnection process. Similarly, the reverse is true if the conventional generator
5 interconnection study process is utilized to enhance requested net injections.

6 IV.F. Interaction with Resource Adequacy Constructs

7 The approach to studying hybrid facilities and their corresponding obligations can interact with
8 resource adequacy studies. We observe that a hybrid facility that can self-supply its energy needs
9 at all times (or otherwise curtail the load) can use that capability to self-supply its resource
10 adequacy requirements so that no resource adequacy responsibilities fall on the rest of the
11 system. A Balancing Authority should recognize this in their resource adequacy studies by not
12 allocating capacity requirements to the load of a hybrid facility and by ensuring that the
13 dispatch/curtailment obligation of the hybrid facilities applies to both resource adequacy and
14 transmission reliability events.

15 IV.G. Ongoing Planning for Operating Limits

16 A holder of executed interconnection agreements of a hybrid facility will not have firm
17 transmission service rights for injections or withdrawals. However, we propose that the
18 maximum operating limit on curtailable net withdrawals of hybrid facilities is fixed in time (with
19 increases possible through network upgrades via the applicable load study process). These non-
20 firm withdrawal operating limits are therefore maintained as necessary in planning studies as the
21 system evolves from year to year. For the operating limit on curtailable net injections of a hybrid
22 facility, we recognize alternative potential policies. The first alternative would permanently fix
23 operating limits on net injections, following the approach for net withdrawals. The second
24 alternative would instead periodically adjust the operating limit (similar to “provisional service”
25 for generators). Under the second alternative, hybrid facilities’ generators face substantial
26 revenue risks because their net injection operating limits can be reduced, potentially to zero and
27 potentially for extended periods of time. On the other hand, under the fixed operating limits of

the first alternative, while revenue risk is lower, future generation planning studies must maintain the net injection operating limits.

V. STREAMLINED STUDY TIMELINE AND DURATION EXPECTATIONS

The streamlined reliability studies and assessments for hybrid facilities include relatively standardized studies that are already in use today (i.e., steady-state power flow, transient stability, short-circuit/breaker duty analysis) as well as studies that are increasingly becoming important with new technologies (e.g., EMT screens and analyses). These studies have been traditionally performed for both generator and load interconnections.

The standard study timeline depends on the operating limit configuration(s) defined in IV.B, the requested facilities that define the footprint of study (i.e., the local area connecting the hybrid facility generation and load), and the number of additional scenarios needed to identify operational impacts of the net grid withdrawals and injections.

When the large load is self-supplied at all times via a behind-the-meter configuration or a front-of-the-meter configuration at the same electrical node (i.e., a zero-megawatt operating limit for both injections and withdrawals), stability and short-circuit analysis can be performed simultaneously and study will focus on a much smaller footprint near the Point of Interconnection (POI). As shown in Appendix I, reliability studies for such cases can generally be completed within 60 days, assuming all information and modeling data is accurate and available from the interconnection application.¹³ Steady-state analysis is not required as the net grid withdrawal and injection is zero megawatt and any inadvertent injection/withdrawals are dealt with operationally (e.g., with protection schemes).

When the hybrid facility's large load is self-supplied at all times via front-of-the-meter generation in a local area configuration (i.e., not at the same electrical node, but electrically close with a zero megawatt operating limit), steady-state analysis, stability, and short-circuit analysis must be performed on the local area network, which is a slightly bigger footprint than the

¹³ Appendix I, Sirisha Tanneeru, et al, "Reliability Assessment of Hybrid Facilities: Overview of Large Load and Generation Co-located Studies", slide 10.

1 behind-the-meter configuration. Nevertheless, it is still likely that these reliability studies can be
2 performed in 60 days. Short-circuit analyses can be performed in parallel with steady-state and
3 stability analyses to streamline the study.

4 Any configuration involving non-zero operating limits for curtailable grid withdrawals or
5 injections requires broader reliability studies and analyses that encompass the broader
6 transmission owner and/or ISO/RTO footprint in addition to the studies required for the local
7 area. These additional studies include steady-state power flow and contingency analysis in
8 addition to transient stability simulations of various load and generation patterns to uncover
9 potential impacts that could be encountered in operations. These studies are used to identify and
10 establish the net withdrawal/injection operating limits for the hybrid facility necessary to avoid
11 critical impacts and identify network upgrades for impacts that cannot be mitigated in operations.
12 Due to their broader reach, the timeline for these studies may exceed 60 days. While most
13 requests can likely be assessed in 60–90 days, as hybrid facility interconnection requests
14 increase, study complexity will rise, which may lead to longer study timelines.

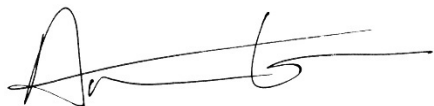
15 Lastly, the stability analyses for hybrid facilities includes a preliminary screen to identify the
16 need for detailed EMT studies. EMT studies are used to identify potential reliability risks, system
17 performance deficiencies, controller interactions, converter instability, weak grid issues, sub-
18 synchronous oscillation risks, and other complex phenomena that cannot be identified using
19 phasor-domain simulation platforms conventionally used in interconnection studies, long-term
20 planning, and operations. Irrespective of the facility's configuration, when the EMT screening
21 results call for comprehensive EMT study, the resulting assessment is resource- and
22 computationally intensive, requires much more detailed and specific data from the
23 interconnection customer, and often requires several months to complete. In situations where
24 EMT studies are identified as necessary, particularly for AI data centers or cryptocurrency loads,
25 these studies should not be overlooked given their criticality in maintaining a reliable bulk power
26 system. When appropriate, EMT studies for multiple hybrid requests could be performed in
27 parallel. Study timelines will exceed 60 days if detailed EMT studies are triggered by the
28 screenings and should be accommodated as such. It is expected that the integration of large
29 electronic loads and the growth of inverter-based generation, each with their unique inverter-
30 based characteristics, will accelerate the need for EMT studies moving forward.

VI. CONCLUSION

Most hybrid facilities can likely be connected without major network upgrades outside their local area when they are curtailable and subject to static operating limits. They can therefore be studied in a rapid, serial, utility-led process focused on identifying the level of fixed operating limits on the net injections and withdrawals from the broader grid that can reliably be accommodated. This meets key objectives of speed to market, associated economic development, and improved outcomes for reliability and affordability. In many cases, such studies can be completed in 60 days for hybrid facilities that self-supply at all times (i.e., zero megawatt operating limit for withdrawals and injections) or in 90 days for other hybrid facilities (e.g., those that request the ability to withdraw power when the grid can accommodate it). For hybrid facilities' interconnection requests that fail EMT screens (e.g., due to weak grid problems) or that are in areas with numerous hybrid facility connection requests, studies may take longer than 90 days but will still benefit from the concepts outlined herein that propose these studies jointly consider both generation and load and the dispatchability of both resources.

Affidavit of Andrew Levitt

I, Andrew Levitt, do hereby swear, that I have co-authored the Affidavit of Andrew Levitt, Aniruddh Mohan, Ryan Quint, and Sirisha Tanneeru, and the statements contained therein are true and accurate to the best of my knowledge and belief.



Andrew Levitt

Date: November 21, 2025.

Affidavit of Aniruddh Mohan

I, Aniruddh Mohan, do hereby swear, that I have co-authored the Affidavit of Andrew Levitt, Aniruddh Mohan, Ryan Quint, and Sirisha Tanneeru, and the statements contained therein are true and accurate to the best of my knowledge and belief.



Aniruddh Mohan

Date: November 21, 2025

Affidavit of Ryan Quint

I, Ryan Quint, do hereby swear, that I have co-authored the Affidavit of Andrew Levitt, Aniruddh Mohan, Ryan Quint, and Sirisha Tanneeru, and the statements contained therein are true and accurate to the best of my knowledge and belief.

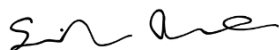


Ryan Quint

Date: November 21, 2025

Affidavit of Sirisha Tanneeru

I, Sirisha Tanneeru, do hereby swear, that I have co-authored the Affidavit of Andrew Levitt, Aniruddh Mohan, Ryan Quint, and Sirisha Tanneeru, and the statements contained therein are true and accurate to the best of my knowledge and belief.



Sirisha Tanneeru

Date: November 21, 2025

Reliability Assessment of Hybrid Facilities

Overview of Large Load and Generation Co-located Studies

Sirisha Tanneeru, Elevate Energy Consulting

November 21, 2025

Kyle Thomas, Elevate Energy Consulting

Andrew Levitt, The Brattle Group



Summary of Reliability Assessment of Hybrid Facilities

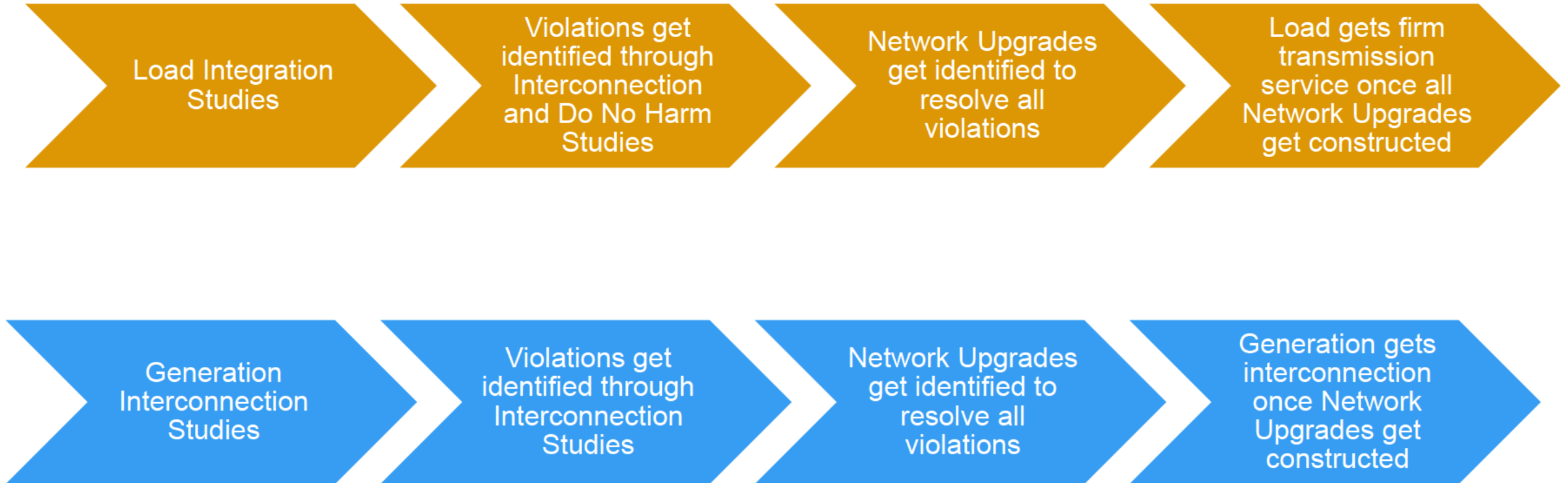
- Hybrid facility (load and near-by generation) obligated to be capable of **self-supply** (i.e., zero net MW) when so dispatched (including load curtailment if near by generation cannot run for any reason)
- Hybrid facility dispatched **independently** by SCED* to ensure N-0 and N-1 flows are within limits
- **No deliverability**: gen is not deliverable, and load does not require deliverability
- **TO conducts independent studies** (like load connection), potentially allowing simpler serial approach

** Unless subject to joint operating limit*

Rapid interconnection is achieved by focusing planning mitigations to only those needed to cover reliability needs not secured in EMS/SCED:

Potential Violations	Potential Mitigations
Short circuit violations	Assume local upgrades needed, not further discussed
Common-mode contingency steady-state violations if not studied in Contingency Analysis (i.e., EMS)	• Post-contingency curtailment via RAS (where practical, potentially as interim measure), <u>or</u> • Pre-contingency curtailment (w/ common-mode outages in Contingency Analysis/SCED)
Post-contingency dynamic stability violations especially for severe, rare common-mode outages (P4/P5/P7); add'l stress-test dispatch cases, 60-90 days adequate. 90+ days in some cases	Combination of: • Joint operating limit on <u>net</u> withdrawal or injection (potentially interim) • RAS (potentially interim) • Network Upgrades

Existing Load & Generation Interconnection Processes Today



Hybrid Facility Co-located Study - High-Level Process

New co-located Interconnection Study

Load Integration
studies occur *with
the addition of the
near-by Generation
included*

Violations get
identified through
the Interconnection
and Do No Harm
Studies

Hybrid facility introduces the
concept of having most of the
violations being mitigated by
Real-time Operations via
SCED

*Possible due to the flexible,
curtailable nature of these
loads

Some hybrid facility violations must
be addressed with conventional
Network Upgrades:

- (1) Ensure Local Area is N-0 and N-1
secure
- (2) Resolve severe violations (e.g.,
instability, breaker duty overloads
etc.) that cannot be resolved in
Real Time Operations via SCED

Electrically Adjacent System “Local Area”

*Hybrid facility Local Area includes load and generation POIs only a few substations apart, confining impacts to the region.
Simple dfax screening can be performed on the TO system to identify if a request qualifies as hybrid facility*

Example – Local Area test criteria

Local Area Test – Example System Intact Criteria TO System up to 3 – 4 levels from Local Area monitored

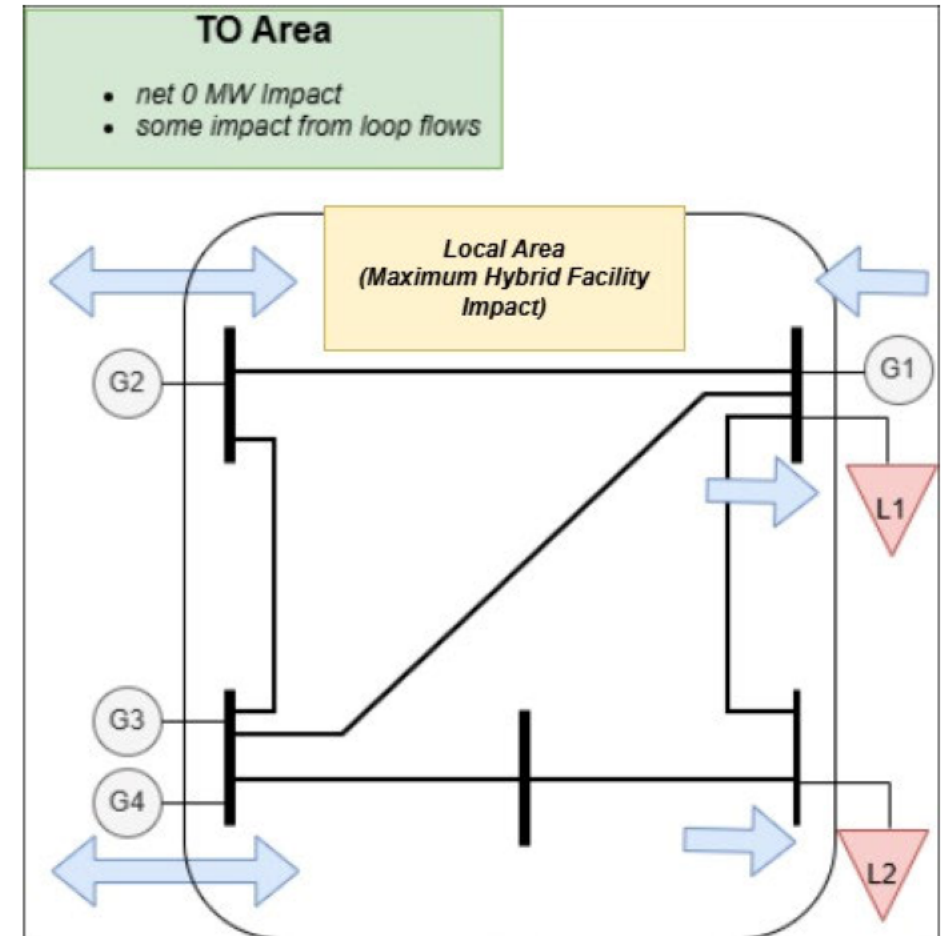
Level 1 and 2 Connections (Local Area ties)

- Dfax of load on lines is <3%; and
- Dfax of generation on lines is <3%

TO Level 3 and 4 Connections

- Sum of all System intact flows on lines is <1% of line rating; and
- Dfax of load on lines is <1%; and
- Dfax of generation on lines is <1%.

Illustrative Picture of Local Area Per SPP Definition



Grid Reliability Challenges & Studies for Large Loads

Common Contributors for Powerflow, Voltage, and Transient Stability Grid Impacts

Transfers
(Imports/ Exports)

Reactive
Power-Voltage
Control

Significant
Gen/Load
Loss

Inadequate
Reactive
power

Ride-Through
Behavior

Common Contributors driving need for EMT Studies

System
Strength

Controller
interactions

Reliability Studies commonly performed for Large Loads (e.g., data centers)

Powerflow
Studies

- Thermal overloads and voltage violations
- Congestion and available transmission services

Transient
Stability
Studies

- Angular stability and electro-mechanical oscillation
- Voltage stability, ride-through performance, etc.

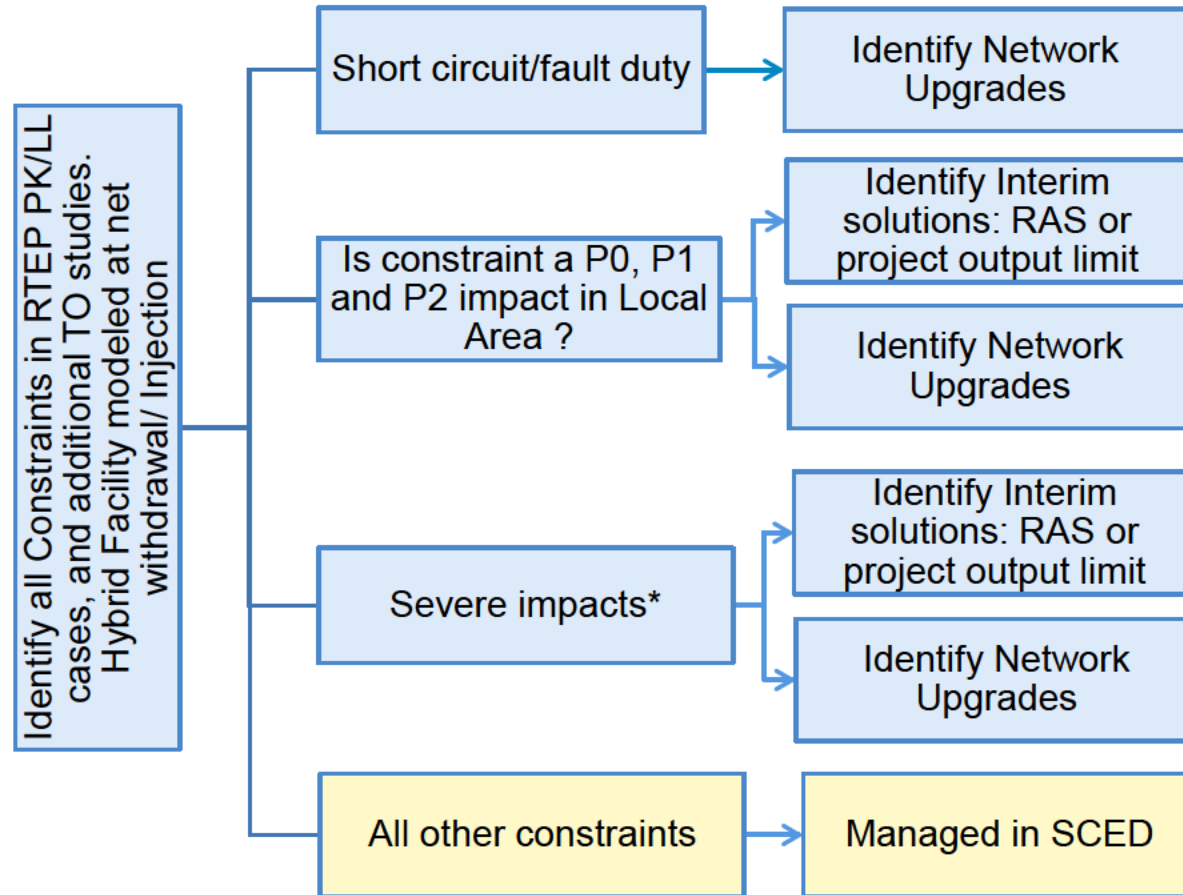
EMT
Studies

- Complex stability phenomena
- Power electronic interactions and oscillations

Short-
Circuit
Studies

- Breaker duty, protection coordination, co-located generation

Hybrid Facility Co-located Interconnection Study Mitigations



Study will model Hybrid Facility gen per the expected output for that fuel type.

Analysis Includes:

- Steady state thermal/voltage
- Stability
- Short circuit/fault duty
- EMT Screening and detailed studies

*Severe Impacts

If impact is severe enough resolving the violations in the Operational timeframe is not feasible, Network Upgrades are identified. Examples may include:

- Thermal overloads encroaching on relay trip limits
- Uncontrolled separation
- Instability
- Large MW load curtailment
- EMT controller interactions
- Breaker over duty limit violations

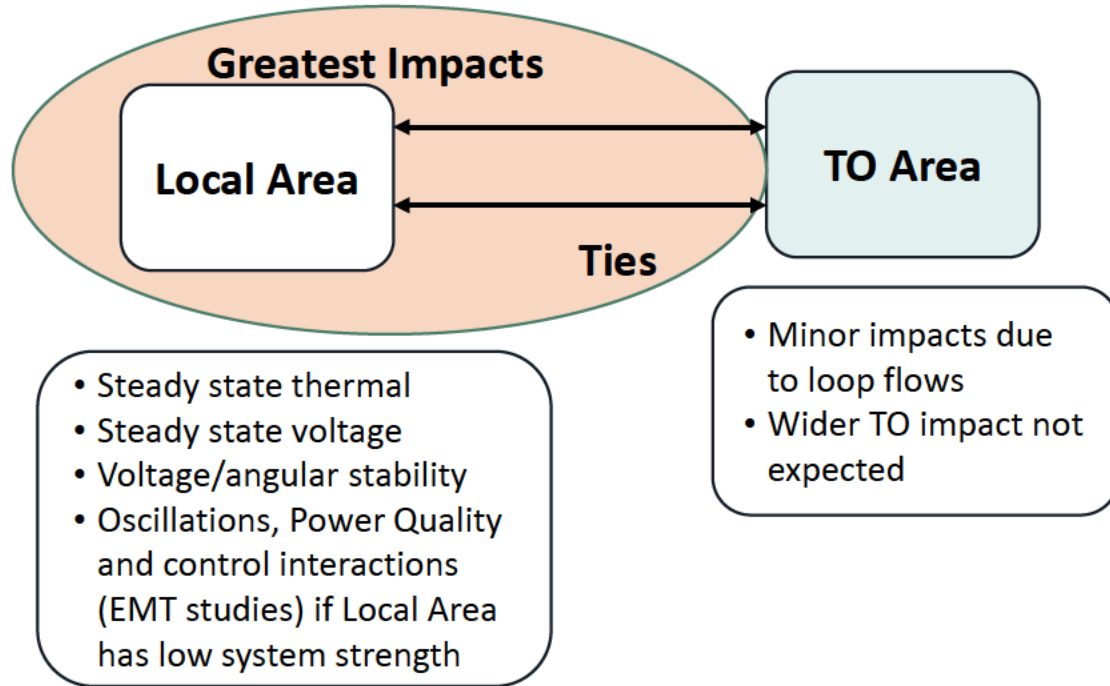
Remedial Action Schemes (RAS)

- Practical for 1–2 busses near POI
- Challenges for wider area implementation
- Requires SCADA/EMS integration
- Operator training needed

Reliability of Large Loads

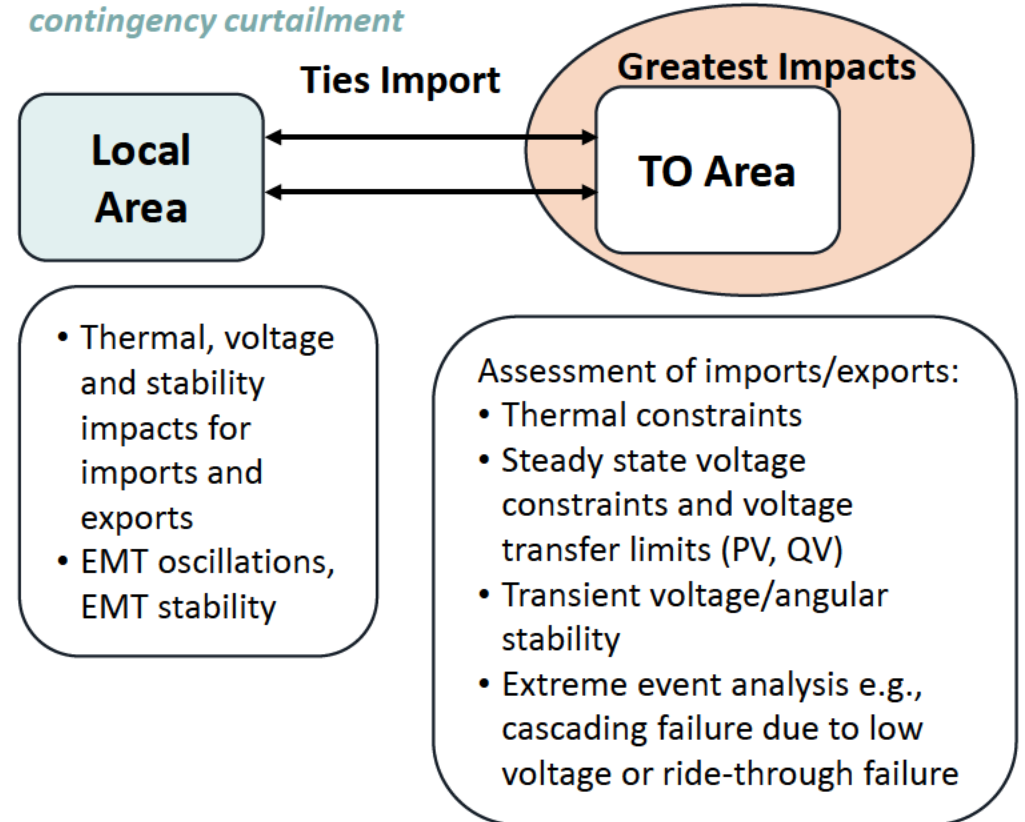
NO GRID WITHDRAWALS

- Local Area most impacted
- Interconnection study will identify all impacts in PK case.
- Local Area is N-0 and N-1 secured. Also, severe constraints that cannot be managed in Operations/SCED are mitigated. Remaining constraints managed by Operations/SECD via load curtailment



GRID WITHDRAWALS and INJECTIONS

- Wider TO area is most impacted. TO shall identify appropriate study case and assumptions to identify potential reliability constraints that could be encountered in SCED
- Interconnection study identifies mitigations for severe contingencies that cannot be managed under SCED
- Analysis in Operational Planning/SCED required to identify pre-contingency curtailment



Potential Operational Planning/SCED Enhancements

- SCED often includes P0 and P1 thermal impact management only. Stability analysis is performed offline in regular intervals or in day-ahead studies performed by operations
- Enhance SCED to include all other contingencies, if possible, otherwise, incorporation of select contingencies impactful to each hybrid facility is recommended. In addition, the enhancements would benefit from the inclusion of analysis for thermal/voltage/stability (to the extent not mitigated in planning)
- Select reliability/congestion limits in operations are nominated by operations and mitigated in the transmission expansion process (status quo)

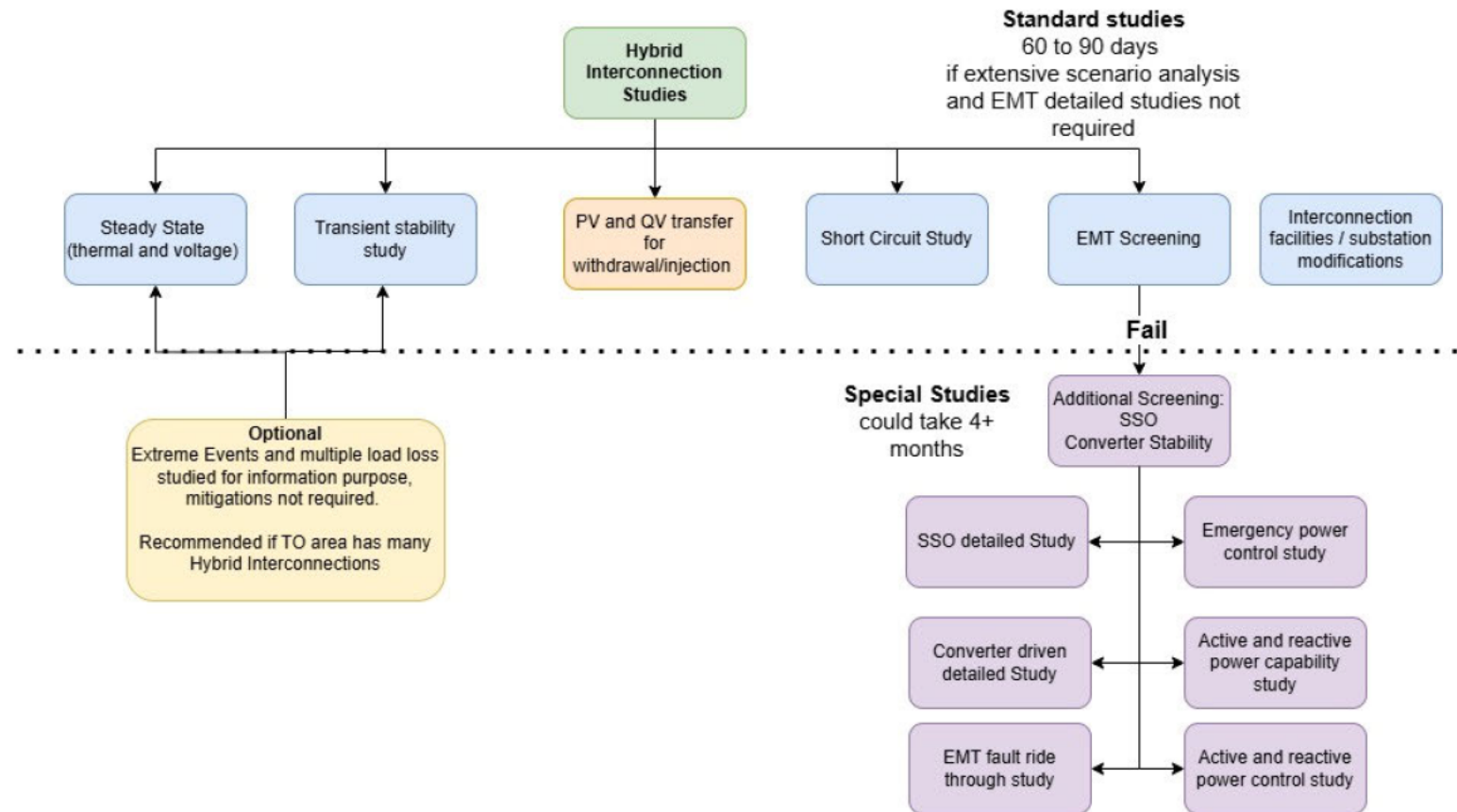
Co-located Interconnection Study – 60 to 90 days

Co-located Interconnection Study can be completed in 60 to 90 days if extensive scenario analysis is not required and EMT screening passes

Serial Study Timeline

- Standard studies could be performed in 60 to 90 days
- In case of failed screen, EMT special studies take 4+ months. Multiple requests in same region can be studied together

Commonly performed studies for Large Load and IBR interconnections



Attachment B

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads	)	Docket No. RM26-4-000
to the Interstate Transmission System	)	

**ATTACHMENT B
COMMENTS OF EOLIAN L.P.
ON THE ADVANCE NOTICE OF PROPOSED RULEMAKING**

List of Key Reforms to Enable Flexible Load and Hybrids

I. Reforms to Load Interconnection Process

- Establish a standardized process, such as a *pro forma* load interconnection process. Ensure key safeguards against undue discrimination applied in the context of the *pro forma* large generator interconnection procedures and agreement, such as the option to build, are incorporated.
- Establish rules regarding queue priority for flexible loads, including defining enforceable commitments to be curtailable (or curtailable and dispatchable for the load study of a hybrid facility) that warrant queue priority.
- Establish rules pursuant to which eligible hybrid facilities may be studied in a separate process, the streamlined study (Category of Reform IV), and avoid duplicative study via the load interconnection process.
- Establish cost allocation rules to apply to load interconnection process.

II. Reforms to Generator Interconnection Queue Process

- Establish rules requiring a study process that incorporates load into a generator interconnection study for generators with defined commitments to serving that load, and corresponding limited interconnection rights to serve that studied load (as opposed to those that would be granted to serve all network loads).
- Establish rules pursuant to which eligible hybrid facilities may be studied in a separate process, the streamlined study (Category of Reform IV), and avoid duplicative study via the generator interconnection process.

III. Reforms to Transmission Service Requirements

- Reform rules to enable non-firm and conditional transmission service other than point-to-point service, including terms of service granting the right of the transmission provider to curtail such loads under particular conditions. Ensure storage resources that are a part of a hybrid resource may be eligible for such new transmission service for charging.
- Reform rules on curtailment of load, as necessary, to reflect the terms of non-firm and conditional transmission service.
- Conforming changes to transmission planning and cost allocation provisions, as necessary, to reflect non-firm and conditional service for relevant loads.

IV. Establish a Streamlined Study Process for Eligible Hybrid Facilities

- Establish a single study, serial interconnection process for hybrid facilities with defined limits on aggregate net injections and withdrawals of power.

V. Reforms to Commission-jurisdictional resource adequacy constructs

- Reforms, as necessary, to recognize the resource adequacy contributions of flexible load and hybrids in existing Commission-jurisdictional resource adequacy constructs.